

Lecture Note

Probability Theory STAT-2101



Faculty of Science
Islamic University
Kushtia, Bangladesh

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Chapter 1

Review of Probability

1.1 References

1. [Mood et al. \(2010\)](#): Probability distributions
2. [Wrede and Spiegel \(2002\)](#): Sequence
3. [Gupta and Kapoor \(2000\)](#): Review of probability, Variate transformation
4. [Sahoo \(2013\)](#): Convergence, Law of large numbers
4. [Roy \(2012\)](#): Probability distributions, Convergence, Law of large numbers
6. Wikipedia: Review of probability, Probability distributions, Convergence, Law of large numbers

1.2 Probability

Probability is a measure quantifying the likelihood that events will occur. Probability quantifies as a number between 0 and 1, where, loosely speaking, 0 indicates impossibility and 1 indicates certainty. The higher the probability of an event, the more likely it is that the event will occur.

These concepts have been given an axiomatic mathematical formalization in probability theory, which is used widely in such areas of study as mathematics, statistics, finance, gambling, science (in particular physics), artificial intelligence/machine learning, computer science, game theory, and philosophy to, for example, draw inferences about the expected frequency of events. Probability theory is also used to describe

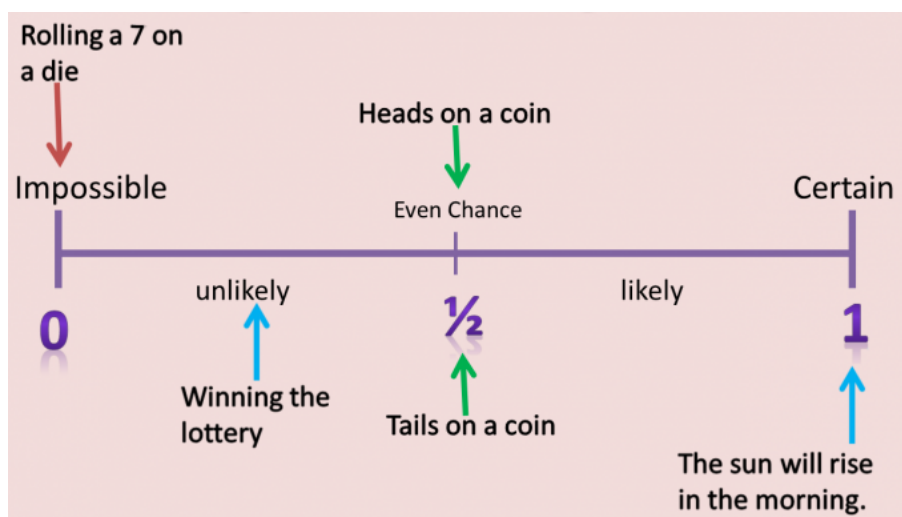


Figure 1.1: The probability scale.

the underlying mechanics and regularities of complex systems.

The probability P of some event E , denoted $P(E)$, is usually defined as to satisfy these axioms. The **Kolmogorov axioms** are described below.

1.3 Axioms

- (i) The probability of an event is a non-negative real number: $P(E) \geq 0 \forall E \in F$, where F is the event space.
- (ii) The probability that at least one of the elementary events in the entire sample space will occur is 1, $P(\Omega) = 1$.
- (iii) Any countable sequence of disjoint sets (synonymous with mutually exclusive events) $E_1, E_2, \dots$ satisfies $P(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$.

1.4 Probability Space

A probability space is a mathematical triplet $(\Omega, \mathcal{F}, P)$ that presents a model for a particular class of real-world situations.

- (i) A sample space, Ω , which is the set of all possible outcomes.

- (ii) The σ -algebra $\mathcal{F} \subseteq 2^\Omega$, is a collection of all the events (not necessarily elementary) we would like to consider.
- (iii) The probability measure P is a function returning an event's probability. Thus P is a function $P : \mathcal{F} \rightarrow [0, 1]$.

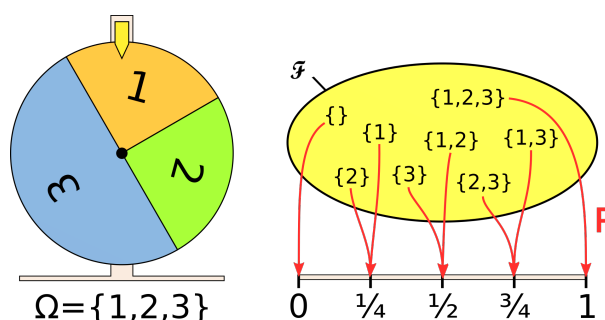


Figure 1.2: Modelling a wheel of fortune using probability space.

1.5 Moment Generating Function

The moment-generating function of a real-valued random variable is an alternative specification of its probability distribution. Thus, it provides the basis of an alternative route to analytical results compared with working directly with probability density functions or cumulative distribution functions. There are particularly simple results for the moment-generating functions of distributions defined by the weighted sums of random variables. However, not all random variables have moment-generating functions. As its name implies, the moment generating function can be used to compute a distribution's moments: the n^{th} moment about 0 is the n^{th} derivative of the moment-generating function, evaluated at 0.

The moment-generating function of a random variable X is

$$M_X(t) = E(e^{tX}), \quad t \in \mathbb{R},$$

wherever this expectation exists. In other words, the moment-generating function is the expectation of the random variable e^{tX} .

The moment-generating function is so named because it can be used to find the moments of the distribution. The series expansion of e^{tX} is

$$e^{tX} = 1 + tX + \frac{t^2}{2!}X^2 + \frac{t^3}{3!}X^3 + \cdots + \frac{t^n}{n!}X^n + \cdots$$

Hence,

$$\begin{aligned} M_X(t) &= E(e^{tX}) = 1 + tE(X) + \frac{t^2}{2!}E(X^2) + \frac{t^3}{3!}E(X^3) + \cdots + \frac{t^n}{n!}E(X^n) + \cdots \\ &= 1 + tm_1 + \frac{t^2}{2!}m_2 + \frac{t^3}{3!}m_3 + \cdots + \frac{t^n}{n!}m_n + \cdots, \end{aligned}$$

where m_n is the n^{th} moment. Differentiating $M_X(t)$, i times with respect to t and setting $t = 0$, we obtain the i^{th} moment about the origin, m_i .

Note: Moment generating function $M_X(t)$ for discrete and continuous random variable can be defined as

$$M_X(t) = E(e^{tX}) = \begin{cases} \sum_{i \in \mathbb{Z}} e^{tx_i} p(x_i), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx & \text{if } X \text{ is continuous,} \end{cases}$$

wherever both of those expectation exists.

1.6 Characteristic Function

The characteristic function of any real-valued random variable completely defines its probability distribution. It provides the basis of an alternative route to analytical results compared with working directly with probability density functions or cumulative distribution functions. There are particularly simple results for the characteristic functions of distributions defined by the weighted sums of random variables.

The characteristic function always exists when treated as a function of a real-valued argument, unlike the moment-generating function.

The characteristic function of a random variable X is

$$\phi_X(t) = E(e^{itX}), \quad t \in \mathbb{R},$$

where $i = \sqrt{-1}$, is the imaginary unit. The characteristic function is the expectation of the random variable e^{itX} . The series expansion of e^{itX} is

$$e^{itX} = 1 + itX + \frac{(it)^2}{2!}X^2 + \frac{(it)^3}{3!}X^3 + \cdots + \frac{(it)^n}{n!}X^n + \cdots.$$

Hence,

$$\begin{aligned} \phi_X(t) &= E(e^{itX}) = 1 + itE(X) + \frac{(it)^2}{2!}E(X^2) + \frac{(it)^3}{3!}E(X^3) + \cdots + \frac{(it)^n}{n!}E(X^n) + \cdots \\ &= 1 + itm_1 + \frac{(it)^2}{2!}m_2 + \frac{(it)^3}{3!}m_3 + \cdots + \frac{(it)^n}{n!}m_n + \cdots, \end{aligned}$$

where m_n is the n^{th} moment. Differentiating $\phi_X(t)$, i times with respect to t and setting $t = 0$, we obtain the i^{th} moment, m_i .

Note: Characteristic generating function $\phi_X(t)$ for discrete and continuous random variable can be defined as

$$\phi_X(t) = E(e^{itX}) = \begin{cases} \sum_{i \in \mathbb{Z}} e^{itx_i} p(x_i), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} e^{itx} f(x) dx & \text{if } X \text{ is continuous,} \end{cases}$$

wherever both of those expectation exists.

1.7 Probability Generating Functions

If X is a discrete random variable taking values in the non-negative integers $\{0, 1, \dots\}$, then the probability generating function of X is defined as

$$G(z) = E(z^X) = \sum_{x=0}^{\infty} p(x)z^x,$$

where p is the probability mass function of X . Note that the subscripted notations G_X and p_X are often used to emphasize that these pertain to a particular random variable X , and to its distribution. The power series converges absolutely at least for all complex numbers z with $|z| \leq 1$; in many examples the radius of convergence is larger.

If $z = e^t$, then this function can be written as

$$G(z) = G(e^t) = E[e^{tX}] = M_X(t).$$

Example 1 Let X be a random variable with probability function

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

Find the probability generating function of X .

Solution: By definition the probability generating function of X is

$$\begin{aligned} G(z) &= \sum_{x=0}^{\infty} z^x p(x) \\ &= \sum_{x=0}^{\infty} z^x \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(z\lambda)^x}{x!} \\ &= e^{-\lambda} e^{z\lambda} = e^{-\lambda(1-z)}, \quad |z| < 1. \end{aligned}$$

Example 2 Let X be a random variable with probability function

$$p(x) = pq^x, \quad x = 0, 1, 2, \dots,$$

such that $0 < p < 1$ and $p + q = 1$. Find the probability generating function of X .

Solution: The probability generating function of X is

$$\begin{aligned}
 G(z) &= \sum_{x=0}^{\infty} z^x p(x) \\
 &= \sum_{x=0}^{\infty} z^x pq^x \\
 &= p \sum_{x=0}^{\infty} (zq)^x \\
 &= p \{1 + zq + (zq)^2 + (zq)^3 + \cdots\} \\
 &= p(1 - zq)^{-1} = \frac{p}{1 - zq}, \quad |z| < 1.
 \end{aligned}$$

1.8 Inversion Theorem

The fundamental theorem of the theory of characteristic function which is known as *Inversion Theorem*, namely that the *characteristic function* uniquely determines the distribution function.

Statement

Let $\phi(t)$ be the characteristic function of a random variable X , then $\phi(t)$ uniquely determines the distribution function $F(x)$ which is

$$F(x) - F(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{-itx}}{it} \phi(t) dt.$$

Further, if $F(x)$ is continuous everywhere and $dF(x) = f(x)dx$, then

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi(t) dt.$$

Proof: For any positive c , let us take the uniformly convergent integral

$$\begin{aligned}
 I_c &= \int_{-c}^c \frac{1 - e^{-itx}}{it} \phi(t) dt \\
 &= \int_{-c}^c \frac{1 - e^{-itx}}{it} \left\{ \int_{-\infty}^{\infty} e^{itu} f(u) du \right\} dt \\
 &= \int_{-c}^c \int_{-\infty}^{\infty} \frac{1 - e^{-itx}}{it} e^{itu} dF(u) dt
 \end{aligned}$$

$$\begin{aligned}
 &= \int_{-c}^c \int_{-\infty}^{\infty} \frac{e^{itu} - e^{it(u-x)}}{it} dF(u) dt \\
 &= \int_{-c}^c \int_{-\infty}^{\infty} \frac{\cos tu + i \sin tu - \cos t(u-x) - i \sin t(u-x)}{it} dF(u) dt \\
 &= \int_{-c}^c \int_{-\infty}^{\infty} \frac{\cos tu - \cos t(u-x)}{it} dF(u) dt \\
 &\quad + \int_{-c}^c \int_{-\infty}^{\infty} \frac{\sin tu - \sin t(u-x)}{t} dF(u) dt.
 \end{aligned} \tag{1.1}$$

Let us take

$$\Psi(t) = \frac{\cos tu - \cos t(u-x)}{it}, \text{ then}$$

$$\begin{aligned}
 \Psi(-t) &= \frac{\cos(-tu) - \cos[-t(u-x)]}{-it} \\
 &= -\frac{\cos tu - \cos t(u-x)}{it} = -\Psi(t).
 \end{aligned}$$

The value of the first integral in (1.1) is 0, since it is an odd function in t .

Now,

$$\begin{aligned}
 I_c &= \int_{-c}^c \int_{-\infty}^{\infty} \frac{\sin tu - \sin t(u-x)}{t} dF(u) dt \\
 &= \int_{-c}^c \int_{-\infty}^{\infty} \frac{\sin tu}{t} dF(u) dt - \int_{-c}^c \int_{-\infty}^{\infty} \frac{\sin t(u-x)}{t} dF(u) dt.
 \end{aligned} \tag{1.2}$$

For any real number β , it is obtained from the Dirichlet integral as

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin t\beta}{t} dt = \operatorname{sgn} \beta,$$

where

$$\operatorname{sgn} \beta = \begin{cases} -1, & \text{if } \beta < 0, \\ 0, & \text{if } \beta = 0, \\ 1, & \text{if } \beta > 0. \end{cases}$$

Thus

$$\int_{-\infty}^{\infty} \frac{\sin tu}{t} dt = \pi \operatorname{sgn} u,$$

and

$$\int_{-\infty}^{\infty} \frac{\sin t(u-x)}{t} dt = \pi \operatorname{sgn} (u-x).$$

Hence, by using (1.2) we get

$$\begin{aligned} I_c &= \int_{-c}^c \pi \operatorname{sgn} u dF(u) - \int_{-c}^c \pi \operatorname{sgn} (u-x) dF(u) \\ \Rightarrow \lim_{c \rightarrow \infty} I_c &= \int_{-\infty}^{\infty} \pi \operatorname{sgn} u dF(u) - \int_{-\infty}^{\infty} \pi \operatorname{sgn} (u-x) dF(u) \\ &= \pi \left[\int_{-\infty}^0 (-1) dF(u) + \int_0^{\infty} (1) dF(u) \right] \\ &\quad - \pi \left[\int_{-\infty}^x (-1) dF(u) + \int_x^{\infty} (1) dF(u) \right] \\ &= \pi [-F(0) + F(-\infty) + F(\infty) - F(0)] \\ &\quad - \pi [-F(x) + F(-\infty) + F(\infty) - F(x)] \\ &= \pi [-F(0) + 0 + 1 - F(0)] - \pi [-F(x) + 0 + 1 - F(x)] \\ &= \pi [-2F(0) + 1] - \pi [-2F(x) + 1] \\ &= 2\pi [F(x) - F(0)]. \end{aligned}$$

Therefore,

$$\begin{aligned} \lim_{c \rightarrow \infty} I_c &= \lim_{c \rightarrow \infty} \int_{-c}^c \frac{1 - e^{-itx}}{it} \phi(t) dt \\ &= 2\pi [F(x) - F(0)]. \end{aligned}$$

Hence,

$$F(x) - F(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{-itx}}{it} \phi(t) dt. \quad (1.3)$$

Now, if we differentiate (1.3) with respect to x , we get

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi(t) dt.$$

This completes the proof of the theorem.

Criterion of Discrete or Continuous Distribution

Let us consider the expression J_c as

$$\begin{aligned} J_c &= \frac{1}{2c} \int_{-c}^c e^{-itx} \phi(t) dt \\ \Rightarrow \lim_{c \rightarrow \infty} J_c &= \lim_{c \rightarrow \infty} \frac{1}{2c} \int_{-c}^c e^{-itx} \phi(t) dt \\ &= 2\pi \lim_{c \rightarrow \infty} \frac{f(x)}{2c} \\ &= 0 \end{aligned}$$

and hence J_c tends to zero at all points where $F(X)$ is continuous and differentiable.

If the distribution function is discontinuous, consider one point of discontinuity, say where the frequency is f_j . The contribution of this part of the frequency to $\phi(t)$ is $f_j e^{itx_j}$ and thus the contribution to J_c is

$$\begin{aligned} J_c &= \frac{1}{2c} \int_{-c}^c e^{-itx} f_j e^{itx_j} dt \\ &= \frac{1}{2c} f_j \int_{-c}^c e^{it(x_j - x)} dt. \end{aligned}$$

If $x_j \neq x$ this clearly tends to 0 as $c \rightarrow \infty$. But if $x_j = x$ the integral is simply

$$J_c = \frac{1}{2c} f_j \int_{-c}^c dt = f_j.$$

Thus J_c tends to f_j at $x = x_j$. Hence if J_c tends to 0 at a point, there is no discontinuity in the distribution function at that point, but if it tends to a positive number f_j the distribution function is discontinuous at that point and the frequency is f_j .

This gives us a criterion where a given characteristic function represents a continuous distribution or not.

Example

Find the density function for which characteristic function is $\phi(t) = e^{-\frac{t^2\sigma^2}{2}}$.

Solution: First, we shall see whether the density function is discrete or continuous.

In this case,

$$\begin{aligned}
 J_c &= \frac{1}{2c} \int_{-c}^c e^{-itx} \phi(t) dt \\
 &= \frac{1}{2c} \int_{-c}^c e^{-itx} e^{-\frac{t^2\sigma^2}{2}} dt \\
 &\leq \frac{1}{2c} \int_{-c}^c e^{-\frac{t^2\sigma^2}{2}} dt ; \quad [e^{-\mathbb{R}} \leq 1] \\
 &\leq \frac{1}{2c} \int_{-\infty}^{\infty} e^{-\frac{t^2\sigma^2}{2}} dt \\
 &= \frac{1}{2c} \frac{\sqrt{2\pi}}{\sigma}.
 \end{aligned}$$

Thus, $\lim_{c \rightarrow \infty} J_c \rightarrow 0$ everywhere. Hence the density function is continuous for all x . So, according to inversion theorem, the probability density function is

$$\begin{aligned}
 f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} e^{-\frac{t^2\sigma^2}{2}} dt \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{t^2\sigma^2}{2} - itx} dt \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(t^2\sigma^2 + 2itx)} dt \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}[(t\sigma)^2 + 2t\sigma \frac{ix}{\sigma} + (\frac{ix}{\sigma})^2 - (\frac{ix}{\sigma})^2]} dt \\
 &= \frac{1}{2\pi} e^{-\frac{x^2}{2\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(t\sigma + \frac{ix}{\sigma})^2} dt \\
 &= \frac{1}{\sigma 2\pi} e^{-\frac{x^2}{2\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz ; \quad \left[t\sigma + \frac{ix}{\sigma} = z \Rightarrow dt = \frac{dz}{\sigma} \right] \\
 &= \frac{1}{\sigma 2\pi} e^{-\frac{x^2}{2\sigma^2}} \sqrt{2\pi} \\
 &= \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}.
 \end{aligned}$$

This is positive everywhere and $\int_{-\infty}^{\infty} f(x)dx = 1$. Therefore, $f(x)$ is a density function. In fact, $f(x)$ is the *pdf* of normal distribution with mean 0 and variance σ^2 .

Example

Find the density function for which characteristic function is $\phi(t) = (q + pe^{it})^n$.

Solution: We have

$$\begin{aligned}\phi(t) &= (q + pe^{it})^n \\ &= \sum_{j=0}^n \binom{n}{j} p^j e^{itj} q^{n-j}.\end{aligned}$$

First, we shall see whether the density function is discrete or continuous. We have

$$\begin{aligned}J_c &= \frac{1}{2c} \int_{-c}^c e^{-itx} \phi(t) dt \\ &= \frac{1}{2c} \int_{-c}^c e^{-itx} \left[\sum_{j=0}^n \binom{n}{j} p^j e^{itj} q^{n-j} \right] dt \\ &= \frac{1}{2c} \sum_{j=0}^n \binom{n}{j} p^j q^{n-j} \int_{-c}^c e^{-it(x-j)} dt\end{aligned}$$

(i) If $x \neq j$, then

$$\begin{aligned}J_c &= \frac{1}{2c} \sum_{j=0}^n \binom{n}{j} p^j q^{n-j} \left[\frac{e^{-it(x-j)}}{-i(x-j)} \right]_{-c}^c \\ &= \frac{1}{2c} \sum_{j=0}^n \binom{n}{j} p^j q^{n-j} \left[\frac{e^{ic(x-j)} - e^{-ic(x-j)}}{i(x-j)} \right] \\ &= \frac{1}{2c} \sum_{j=0}^n \binom{n}{j} p^j q^{n-j} \left[\frac{2i \sin c(x-j)}{i(x-j)} \right] ; \quad [e^{\pm ix} = \cos x \pm i \sin x]\end{aligned}$$

$\therefore \lim_{c \rightarrow \infty} J_c \rightarrow 0$ for all x .

(ii) If $x = j$, then

$$J_c = \frac{1}{2c} \sum_{j=0}^n \binom{n}{j} p^j q^{n-j} \int_{-c}^c dt$$

$$\begin{aligned}
 &= \sum_{j=0}^n \binom{n}{j} p^j q^{n-j} \\
 &= 1.
 \end{aligned}$$

Thus $J_c = 1$ when $x = j$. Therefore, the density function is discrete and its probability function is

$$p(x) = \binom{n}{j} p^j q^{n-j}, \quad j = 0, 1, 2, \dots, n.$$

Example

Find the density function for which characteristic function is $\phi(t) = e^{-|t|}$.

Solution: We have

$$J_c = \frac{1}{2c} \int_{-c}^c e^{-itx} e^{-|t|} dt. \quad (1.4)$$

It can be shown that $\lim_{c \rightarrow \infty} J_c \rightarrow 0$. Hence the distribution is continuous. So the density function is

$$\begin{aligned}
 f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} e^{-|t|} dt \\
 &= \frac{1}{2\pi} \int_{-\infty}^0 e^{t-ix} dt + \frac{1}{2\pi} \int_0^{\infty} e^{-t-ix} dt \\
 &= \frac{1}{2\pi} \int_0^{\infty} e^{-t+itx} dt + \frac{1}{2\pi} \int_0^{\infty} e^{-t-ix} dt \\
 &= \frac{1}{2\pi} \left[\int_0^{\infty} e^{-t} (e^{itx} + e^{-itx}) dt \right] \\
 \Rightarrow f(x) &= \frac{1}{\pi} \int_0^{\infty} e^{-t} \cos tx \, dt \\
 &= \frac{1}{2\pi} \left[\int_0^{\infty} e^{-t} (e^{itx} + e^{-itx}) dt \right] \\
 &= \frac{1}{\pi} \int_0^{\infty} \cos tx \, e^{-t} dt \\
 &= \frac{1}{\pi} [-e^{-t} \cos tx]_0^{\infty} - \frac{x}{\pi} \int_0^{\infty} e^{-t} \sin tx \, dt
 \end{aligned} \quad (1.5)$$

$$\begin{aligned}
 &= \frac{1}{\pi} + \frac{x}{\pi} [e^{-t} \sin tx]_0^\infty - \frac{x^2}{\pi} \int_0^\infty e^{-t} \cos tx \, dt \\
 &= \frac{1}{\pi} - x^2 f(x).
 \end{aligned}$$

Therefore

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

In fact, this is the well-known standard Cauchy distribution.

Assignment: Find the density function for which characteristic function is

$$\phi(t) = \begin{cases} 1 - |t|, & |t| \leq 1, \\ 0, & |t| \geq 1. \end{cases}$$

1.9 Laplace Transformation and its Properties

1.9.1 Laplace Transformation

Let $F(t)$ be a function of t specified for $t > 0$. Then the Laplace transform of $F(t)$, denoted by $\mathcal{L}\{F(t)\}$, is defined by

$$\mathcal{L}\{F(t)\} = f(s) = \int_0^\infty e^{-st} F(t) dt \quad (1.6)$$

where we assume at present that the parameter s is real. Later it will be found useful to consider s complex. The Laplace transform of $F(t)$ is said to exist if the integral (1.6) converges for some value of s ; otherwise it does not exist.

Laplace Transformation of Some Elementary Functions

Prove that:

$$(i) \quad \mathcal{L}\{1\} = \frac{1}{s}, \quad s > 0;$$

$$(ii) \mathcal{L}\{t\} = \frac{1}{s^2}, \quad s > 0;$$

$$(iii) \mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad s > a;$$

Proof:

(a)

$$\begin{aligned} \mathcal{L}\{F(t)\} &= f(s) = \int_0^\infty e^{-st} F(t) dt \\ \Rightarrow \mathcal{L}\{1\} &= f(s) = \int_0^\infty e^{-st} (1) dt \\ &= \lim_{P \rightarrow \infty} \int_0^P e^{-st} dt \\ &= \lim_{P \rightarrow \infty} \left. \frac{e^{-st}}{-s} \right|_0^P \\ &= \lim_{P \rightarrow \infty} \frac{1 - e^{-sP}}{s} \\ &= \frac{1}{s} \quad \text{if } s > 0. \end{aligned}$$

(b)

$$\begin{aligned} \mathcal{L}\{F(t)\} &= f(s) = \int_0^\infty e^{-st} F(t) dt \\ \Rightarrow \mathcal{L}\{t\} &= f(s) = \int_0^\infty e^{-st} (t) dt \\ &= \int_0^\infty t e^{-st} dt \quad \left[\int u v dx = u \int v dx - \int u' \left(\int v dx \right) dx \right] \\ &= \lim_{P \rightarrow \infty} \left\{ (t) \left(\frac{e^{-st}}{-s} \right) - (1) \left(\frac{e^{-st}}{s^2} \right) \right\} \Big|_0^P \\ &= \lim_{P \rightarrow \infty} \left\{ (P) \left(\frac{e^{-sP}}{-s} \right) - \left(\frac{e^{-sP}}{s^2} \right) + \left(\frac{1}{s^2} \right) \right\} \\ &= \frac{1}{s^2} \quad \text{if } s > 0. \end{aligned}$$

(c)

$$\begin{aligned} \mathcal{L}\{F(t)\} &= f(s) = \int_0^\infty e^{-st} F(t) dt \\ \Rightarrow \mathcal{L}\{e^{at}\} &= f(s) = \int_0^\infty e^{-st} (e^{at}) dt \end{aligned}$$

$$\begin{aligned}
 &= \lim_{P \rightarrow \infty} \int_0^P e^{-(s-a)t} dt \\
 &= \lim_{P \rightarrow \infty} \left. \frac{e^{-(s-a)t}}{-(s-a)} \right|_0^P \\
 &= \lim_{P \rightarrow \infty} \frac{1 - e^{-(s-a)P}}{s-a} \\
 &= \frac{1}{s-a} \quad \text{if } s > a.
 \end{aligned}$$

Assignment: Prove that:

(i) $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}, \quad \text{if } s > 0;$

(ii) $\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}, \quad \text{if } s > 0.$

Assignment:

Find $\mathcal{L}\{F(t)\}$ if $F(t) = \begin{cases} 5 & \text{if } 0 < t < 3 \\ 0 & \text{if } t > 3 \end{cases}$.

Hint's: $\mathcal{L}\{F(t)\} = \frac{5(1-e^{-3s})}{s}$

1.9.2 Properties of Laplace Transformation

- **Linearity property:** $\mathcal{L}\{c_1 F_1(t) + c_2 F_2(t)\} = \mathcal{L}\{c_1 F_1(t)\} + \mathcal{L}\{c_2 F_2(t)\}$
- **Shifting property:** $\mathcal{L}\{e^{at} F(t)\} = f(s-a)$
- **Change of scale property:** $\mathcal{L}\{F(at)\} = \frac{1}{a} f\left(\frac{s}{a}\right)$
- **Laplace transformation of derivatives:** $\mathcal{L}\{F'(t)\} = sf(s) - F(0)$
- **Laplace transformation of integrals:** $\mathcal{L}\left\{\int_0^t F(u)du\right\} = \frac{f(s)}{s}$
- **Multiplication by t^n :** $\mathcal{L}\{t^n F(t)\} = (-1)^n f^{(n)}(s)$
- **Division by t :** $\mathcal{L}\left\{\frac{F(t)}{t}\right\} = \int_s^\infty F(u)du$
- **Initial-value theorem** $\lim_{t \rightarrow 0} F(t) = \lim_{s \rightarrow \infty} sf(s)$

- **Final-value theorem** $\lim_{t \rightarrow \infty} F(t) = \lim_{s \rightarrow 0} s f(s)$

****Assignment:** State and prove the properties of Laplace transformation.

***Assignment:** Write down the applications of Laplace transformation.

1.10 Convolution

Convolution is a mathematical operation on two functions (f and g) to produce a third function that expresses how the shape of one is modified by the other.

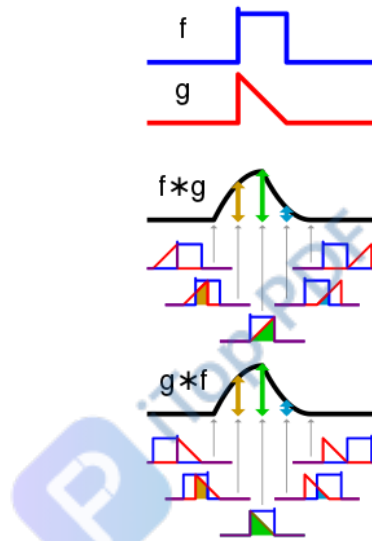


Figure 1.3: The convolution

We illustrate how convolution technique can be used in finding the distribution of the sum of random variables when they are independent. *This convolution technique does not work if the random variables are not independent.*

Definition:

Let f and g be two real valued functions. The convolution of f and g , denoted by $f * g$, is defined as

$$\begin{aligned} (f * g)(z) &= \int_{-\infty}^{\infty} f(z - y)f(y)dy \\ &= \int_{-\infty}^{\infty} f(z - x)f(x)dy \end{aligned}$$

Hence from this definition it is clear that $f * g = g * f$.

Let X and Y be two independent random variables with probability density functions $f(x)$ and $g(y)$. Here, we get

$$h(z) = \int_{-\infty}^{\infty} f(z-y)f(y)dy.$$

Thus, this result shows that the density of the random variable $Z = X + Y$ is the convolution of the density of X with the density of Y .

Applications: Convolution has applications that include probability, statistics, computer vision, natural language processing, image and signal processing, engineering, and differential equations.

Example

If x_1 and x_2 are independent $N(0, 1)$ variates, find the distribution of $z = x_1 + x_2$.

Solution: Here x_1 and x_2 are independent $N(0, 1)$ variates, so

$$\begin{aligned} h(z) &= (f_1 * f_2)(z) = \int_{-\infty}^{\infty} f_1(z-x_2)f_2(x_2)dx_2 \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-x_2)^2} e^{-\frac{1}{2}x_2^2} dx_2 \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\left\{\frac{z^2}{2}-zx_2+x_2^2\right\}} dx_2 \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\left\{x_2^2-zx_2+\frac{z^2}{4}+\frac{z^2}{4}\right\}} dx_2 \\ &= \frac{1}{2\pi} e^{-\frac{z^2}{4}} \int_{-\infty}^{\infty} e^{-\left\{x_2-\frac{z}{2}\right\}^2} dx_2 \\ &= \frac{1}{2\pi} e^{-\frac{z^2}{4}} \int_{-\infty}^{\infty} e^{-w^2} dw \\ &= \frac{1}{\sqrt{2}\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{z-0}{\sqrt{2}}\right)^2} \end{aligned}$$

Hence, z is an $N(0, 2)$ variate.

Example

What is the probability density of the sum of two independent random variables, each of which is $GAM(1, 1)$ distribution?

Solution: Let X and Y be two random variables such that $X, Y \sim GAM(1, 1)$. Hence, the density functions $f(x)$ and $f(y)$ can be written as

$$\begin{aligned} f(x) &= e^{-x} \quad \text{for } 0 < x < \infty, \quad \text{and} \\ f(y) &= e^{-y} \quad \text{for } 0 < y < \infty. \end{aligned}$$

Let $Z = X + Y$. Since X and Y are independent, the density function of Z can be obtained by the method of convolution.

Notice that the sum $z = x + y$ is between 0 and 1, and $0 < x < z$. Hence, the density function of Z is given by

$$\begin{aligned} h(z) &= (f * g)(z) \\ &= \int_{-\infty}^{\infty} f(z-x) f(x) dx \\ &= \int_0^{\infty} f(z-x) f(x) dx \\ &= \int_0^z e^{-(z-x)} e^{-x} dx \\ &= \int_0^z e^{-z} dx \\ &= ze^{-z} \\ &= \frac{1^2 e^{-z} z^{2-1}}{\Gamma(2)}. \end{aligned}$$

Hence $Z \sim GAM(1, 2)$.

Chapter 2

Probability Distributions

The probability distribution is a mathematical function that provides the probabilities of occurrence of different possible outcomes in an experiment. In more technical terms, the probability distribution is a description of a random phenomenon in terms of the probabilities of events.

For instance, if the random variable X is used to denote the outcome of a coin toss ("the experiment"), then the probability distribution of X would take the value 0.5 for $X = heads$, and 0.5 for $X = tails$ (assuming the coin is fair).

Probability distributions are generally divided into two classes. A discrete probability distribution (applicable to the scenarios where the set of possible outcomes is discrete, such as a coin toss or a roll of dice) can be encoded by a discrete list of the probabilities of the outcomes, known as a probability mass function. On the other hand, a continuous probability distribution (applicable to the scenarios where the set of possible outcomes can take on values in a continuous range (e.g. real numbers), such as the temperature on a given day) is typically described by probability density functions (with the probability of any individual outcome actually being 0).

2.1 Laplace Distribution

The Laplace distribution is a continuous probability distribution named after **Pierre-Simon Laplace**. It is also sometimes called the double exponential distribution,

because it can be thought of as two exponential distributions (with an additional location parameter) spliced together.

2.1.1 Standard Laplace Distribution

A continuous random variable X is said to follow standard Laplace distribution if its *p.d.f.* is given by

$$f(x) = \frac{1}{2}e^{-|x|}; \quad -\infty < x < \infty.$$

2.1.1.1 Characteristic Function

The characteristic function of standard distribution can be written as

$$\begin{aligned} \phi_X(t) &= \int_{-\infty}^{\infty} e^{itx} f(x) dx \\ &= \frac{1}{2} \int_{-\infty}^{\infty} e^{itx} e^{-|x|} dx \\ &= \frac{1}{2} \left[\int_{-\infty}^{\infty} \cos(tx) e^{-|x|} dx + i \int_{-\infty}^{\infty} \sin(tx) e^{-|x|} dx \right] \\ &= \int_0^{\infty} \cos(tx) e^{-|x|} dx, \end{aligned}$$

since the integrands in the first and second integrals are even and odd function of x respectively. Now further obtain as

$$\begin{aligned} \phi_X(t) &= \int_0^{\infty} e^{-x} \cos(tx) dx, \\ &= e^{-x} \frac{1}{t} \sin(tx) \Big|_0^{\infty} - \int_0^{\infty} (-e^{-x}) \frac{1}{t} \sin(tx) dx \\ &= (0 - 0) - e^{-x} \frac{1}{t^2} \cos(tx) \Big|_0^{\infty} + \frac{1}{t} \left[(-e^{-x}) \frac{1}{t} \cos(tx) \Big|_0^{\infty} - \int_0^{\infty} e^{-x} \frac{1}{t} \cos(tx) dx \right] \\ &= \frac{1}{t^2} - \frac{\phi_X(t)}{t^2} \\ &= \frac{1}{1 + t^2} \\ &= (1 + t^2)^{-1} \\ &= 1 - t^2 + t^4 - t^6 + \dots \end{aligned}$$

$$= 1 + 2\frac{(it)^2}{2!} + 4!\frac{(it)^4}{4!} - \dots \quad (2.1)$$

2.1.1.2 Moments

The r_{th} raw moments is the coefficients of the term $\frac{(it)^r}{r!}$ in equation (2.1), and can be easily obtained as follows

$$\begin{aligned} \mu'_1 &= \mu'_3 = 0, \\ \mu'_2 &= 2, \\ \mu'_4 &= 4! = 24, \\ Mean &= \mu'_1 = 0, \\ Variance &= \mu_2 = \mu'_2 - \mu'^2_1 = 2, \\ \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2\mu'^3_1 = 0, \\ \mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu'^2_1 - 3\mu'^4_1 = 24, \\ \beta_1 &= \frac{\mu^2_3}{\mu^3_2} = 0 \\ \beta_2 &= \frac{\mu_4}{\mu^2_2} = 6. \end{aligned}$$

2.1.2 Two Parameter Laplace Distribution

A continuous random variable X is said to have a double exponential (Laplace) distribution with two parameters λ and μ if its *p.d.f.* is given by

$$f(x; \mu, \sigma) = \frac{1}{2\lambda} e^{-\frac{|x-\mu|}{\lambda}}; \quad -\infty < x < \infty, \quad \lambda > 0.$$

We write $X \sim Lap(\lambda, \mu)$. Let $Y = \frac{X-\mu}{\lambda} \Rightarrow X = \mu + \lambda Y$. The *p.d.f.* $g(\cdot)$ of Y is given as follows:

$$\begin{aligned} g(y) &= f(x) \left| \frac{dx}{dy} \right| = \frac{1}{2\lambda} e^{-|y|} \lambda \\ &= \frac{1}{2} e^{-|y|}; \quad -\infty < y < \infty, \end{aligned} \quad (2.2)$$

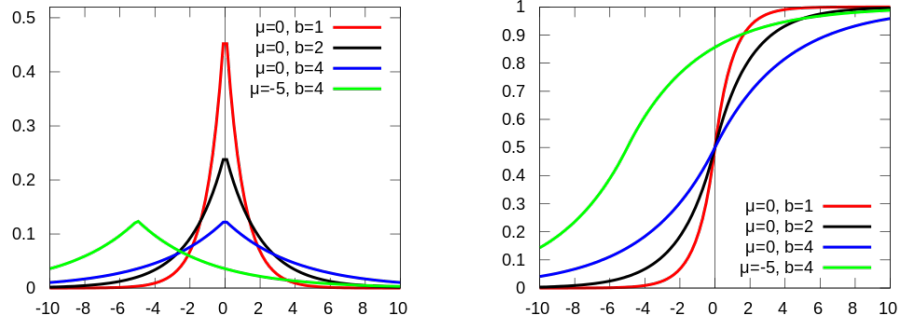


Figure 2.1: The *pdf* and *cdf* of Laplace distribution.

which is the *p.d.f.* of standard Laplace distribution.

Hence if $X \sim \text{Lap}(\lambda, \mu)$, then $Y = \frac{X - \mu}{\lambda} \sim \text{Lap}(1, 0)$, i.e., Y has the standard Laplace distribution.

2.1.2.1 Characteristic Function

If $X \sim \text{Lap}(\lambda, \mu)$, then

$$\begin{aligned}
 \phi_X(t) &= E(e^{itx}) \\
 &= E[e^{it(\mu + \lambda Y)}], \quad Y = \frac{X - \mu}{\lambda} \sim \text{Lap}(1, 0), \\
 &= e^{it\mu} E(e^{i\lambda t Y}) \\
 &= e^{it\mu} \phi_Y(\lambda t) \\
 &= \frac{e^{it\mu}}{1 + \lambda^2 t^2}
 \end{aligned}$$

Here, Y is standard Laplace variate with $\phi(t) = \frac{1}{1+t^2}$.

2.1.2.2 Moments

If $X \sim \text{Lap}(\lambda, \mu)$, then r^{th} moment about origin is given by:

$$\begin{aligned}
 \mu'_r &= \frac{1}{2\lambda} \int_{-\infty}^{\infty} x^r e^{-\frac{|x-\mu|}{\lambda}} dx \\
 &= \frac{1}{2} \int_{-\infty}^{\infty} (\mu + z\lambda)^r e^{-|z|} dz; \quad \left[z = \frac{x - \mu}{\lambda} \Rightarrow dx = \lambda dz \right]
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \int_{-\infty}^{\infty} \left[\sum_{k=0}^r \binom{r}{k} (z\lambda)^k \mu^{r-k} \right] e^{-|z|} dz \\
&= \frac{1}{2} \sum_{k=0}^r \left[\binom{r}{k} (\lambda)^k \mu^{r-k} \int_{-\infty}^{\infty} z^k e^{-|z|} dz \right] \\
&= \frac{1}{2} \sum_{k=0}^r \left[\binom{r}{k} (\lambda)^k \mu^{r-k} \left\{ \int_{-\infty}^0 z^k e^{-|z|} dz + \int_0^{\infty} z^k e^{-|z|} dz \right\} \right] \\
&= \frac{1}{2} \sum_{k=0}^r \left[\binom{r}{k} (\lambda)^k \mu^{r-k} \left\{ (-1)^k \int_0^{\infty} e^{-z} z^k dz + \int_0^{\infty} e^{-z} z^k dz \right\} \right] \\
&= \frac{1}{2} \sum_{k=0}^r \left[\binom{r}{k} (\lambda)^k \mu^{r-k} \Gamma(k+1) \{(-1)^k + 1\} \right] \\
&= \frac{1}{2} \sum_{k=0}^r \left[\binom{r}{k} (\lambda)^k \mu^{r-k} k! \{1 + (-1)^k\} \right]. \tag{2.3}
\end{aligned}$$

$\therefore$ Mean $= \mu'_1 = \mu$, $\mu'_2 = (\mu)^2 + (0) + (2\lambda^2)$ and Variance $= \mu_2 = \mu'_2 - \mu_1'^2 = 2\lambda^2$.

Similarly we can obtain higher order moments from (2.3) and hence the values of β_1 and β_2 can be obtained.

2.1.3 Applications

The Laplace distribution is used for modeling in signal processing, speech recognition, various biological processes, finance, hydrology, regression analysis, and economics.

2.2 Cauchy Distribution

The Cauchy distribution, named after **Augustin Cauchy**, is a continuous probability distribution. It is also known as the Lorentz distribution (after Hendrik Lorentz), Cauchy–Lorentz distribution, Lorentz(ian) function, or Breit–Wigner distribution.

Definition:

A random variable X is said to have a standard Cauchy distribution if its probability

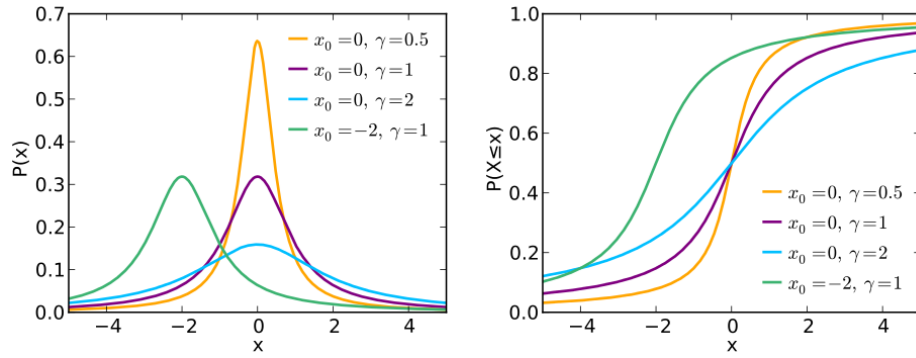


Figure 2.2: The *pdf* and *cdf* of Cauchy distribution.

density function (*p.d.f.*) is given by

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty, \quad (2.4)$$

and X is termed as standard Cauchy variate.

More generally, Cauchy distribution with parameters λ and μ has the *p.d.f.*

$$g_Y(y) = \frac{\lambda}{\pi[\lambda^2 + (y - \mu)^2]}, \quad -\infty < x < \infty; \lambda > 0, \quad (2.5)$$

and we can write $Y \sim C(\lambda, \mu)$.

But putting $X = \frac{Y-\mu}{\lambda}$ in (2.5), we get (2.4).

Hence if $Y \sim C(\lambda, \mu)$, then $X = \frac{Y-\mu}{\lambda} \sim C(1, 0)$.

2.2.1 Moments

Moments of Cauchy distribution can be obtain as

$$\begin{aligned} F(Y) &= \int_{-\infty}^{\infty} y f(y) dy \\ &= \int_{-\infty}^{\infty} y \frac{\lambda}{\pi[\lambda^2 + (y - \mu)^2]} dy \end{aligned}$$

$$\begin{aligned}
 &= \frac{\lambda}{\pi} \int_{-\infty}^{\infty} \frac{(y - \mu) + \mu}{[\lambda^2 + (y - \mu)^2]} dy \\
 &= \mu \frac{\lambda}{\pi} \int_{-\infty}^{\infty} \frac{dy}{[\lambda^2 + (y - \mu)^2]} + \frac{\lambda}{\pi} \int_{-\infty}^{\infty} \frac{(y - \mu)}{[\lambda^2 + (y - \mu)^2]} dy \\
 &= \mu + \frac{\lambda}{\pi} \int_{-\infty}^{\infty} \frac{z}{\lambda^2 + z^2} dz \quad ; [y - \mu = z \Rightarrow dy = dz].
 \end{aligned}$$

Although the integral $\int_{-\infty}^{\infty} \frac{z}{\lambda^2 + z^2} dz$, is not completely convergent. Thus, in the general sense the mean of Cauchy distribution does not exist. But, if we conventionally agree to assume that the mean of Cauchy distribution exists (by taking the principal value), then it is located at $x = \mu$. Also, obviously, the probability curve is symmetrical about the point $x = \mu$. Hence for this distribution, the mean, median and mode coincide at the point $x = \mu$. Similarly

$$\begin{aligned}
 \mu_2 &= E(Y - \mu)^2 \\
 &= \int_{-\infty}^{\infty} (y - \mu)^2 f(y) dy \\
 &= \frac{\lambda}{\pi} \int_{-\infty}^{\infty} \frac{(y - \mu)^2}{\lambda^2 + (y - \mu)^2} dy,
 \end{aligned}$$

which does not exist since the integral is not convergent. Thus, in general, for the Cauchy's distribution, the moments μ_r , ($r \geq 2$) do not exist.

2.2.2 Characteristic Function

If X is a standard Cauchy variate then

$$\phi_X(t) = \int_{-\infty}^{\infty} \frac{e^{itx}}{\pi(1+x^2)} dx \quad (2.6)$$

To evaluate (2.6) consider standard Laplace distribution

$$f_1(z) = \frac{1}{2} e^{-|z|}, \quad -\infty < z < \infty.$$

Then

$$\phi_1(t) = \phi_Z(t) = E(e^{itz}) = \frac{1}{1+t^2}.$$

Since $\phi_1(t)$ is absolutely integrable in $(-\infty, \infty)$, we have by Inversion theorem

$$\begin{aligned} \frac{1}{2}e^{-|z|} &= f_1(z) \\ \Rightarrow \frac{1}{2}e^{-|z|} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itz} \phi_1(t) dt \\ \Rightarrow \frac{1}{2}e^{-|z|} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{-itz}}{1+t^2} dt \\ \therefore e^{-|z|} &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{itz}}{1+t^2} dt \quad ; [\text{Changing } t \text{ to } -t]. \end{aligned}$$

On interchanging t and z , we have

$$e^{-|t|} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{itz}}{1+z^2} dz. \quad (2.7)$$

From (2.6) and (2.2.2), we get $\phi_X(t) = e^{-|t|}$.

Now if Y is a Cauchy variate with parameters λ and μ , then

$$X = \frac{Y - \mu}{\lambda} \sim C(1, 0) \Rightarrow Y = \mu + \lambda X.$$

Therefore

$$\begin{aligned} \phi_Y(t) &= E(e^{itY}) \\ &= e^{it\mu} E(e^{it\lambda X}) \\ &= e^{it\mu} \phi_X(t\lambda) \\ &= e^{it\mu - \lambda|t|}, \quad \lambda > 0. \end{aligned}$$

2.2.3 Additive Property

If X_1 and X_2 are independent Cauchy variates with parameters (λ_1, μ_1) and (λ_2, μ_2) respectively, then $X_1 + X_2$ is a Cauchy variate with parameters $(\lambda_1 + \lambda_2, \mu_1 + \mu_2)$.

Proof: We have

$$\begin{aligned}\phi_{X_i}(t) &= e^{it\mu_i - \lambda_i|t|}, \quad (i = 1, 2) \\ \therefore \phi_{X_1+X_2}(t) &= \phi_{X_1}(t) \cdot \phi_{X_2}(t) \\ &= e^{it\mu_1 - \lambda_1|t|} \cdot e^{it\mu_2 - \lambda_2|t|} \\ &= e^{it(\mu_1 + \mu_2) - (\lambda_1 + \lambda_2)|t|}\end{aligned}$$

and the result follows by uniqueness theorem of characteristic functions.

Example

Let X have a standard Cauchy distribution. Find the probability density function of X^2 and identify its distribution.

Solution: Since X has a standard Cauchy distribution, its probability density function can be written as

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

The distribution function of $Y = X^2$ is

$$\begin{aligned}G_Y(y) &= P(Y \leq y) \\ &= P(X^2 \leq y) \\ &= P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} f(x) dx\end{aligned}$$

$$\begin{aligned}
 &= \frac{2}{\pi} \int_0^{\sqrt{y}} \frac{1}{1+x^2} dx \\
 &= \frac{2}{\pi} \tan^{-1}(\sqrt{y}), \quad 0 < y < \infty.
 \end{aligned}
 \tag{2.8}$$

The density function of Y is

$$\begin{aligned}
 g_Y(y) &= \frac{d}{dy} [G_Y(y)] \\
 &= \frac{2}{\pi} \cdot \frac{1}{1+y} \cdot \frac{1}{2\sqrt{y}} \\
 &= \frac{1}{\beta\left(\frac{1}{2}, \frac{1}{2}\right)} \frac{y^{\frac{1}{2}-1}}{(1+y)^{\frac{1}{2}+\frac{1}{2}}}, \quad y > 0.
 \end{aligned}
 \tag{2.9}$$

This is the density function of Beta distribution of second kind with parameters $\left(\frac{1}{2}, \frac{1}{2}\right)$, i.e., $X^2 \sim \beta_2\left(\frac{1}{2}, \frac{1}{2}\right)$.

Assignment

Let $X \sim N(0, 1)$ and $Y \sim N(0, 1)$ be independent random variables. Find the distribution of $\frac{X}{Y}$ and identify it.

2.2.4 Applications

Applications of the Cauchy distribution or its transformation can be found in fields working with exponential growth. Outside of finance, the Cauchy distribution is often the distribution of observations for objects that are spinning. In hydrology the Cauchy distribution is applied to extreme events such as annual maximum one-day rainfalls and river discharges.

2.3 Logistic Distribution

The logistic distribution is a continuous probability distribution. Its cumulative distribution function is the logistic function, which appears in logistic regression and

feedforward neural networks. It resembles the normal distribution in shape but has heavier tails (higher kurtosis). The logistic distribution is a special case of the Tukey lambda distribution.

2.3.1 Definition

The probability density function of the logistic distribution is given by:

$$f(x; \alpha, \beta) = \frac{1}{\beta} \frac{e^{-\frac{x-\alpha}{\beta}}}{\left(1 + e^{-\frac{x-\alpha}{\beta}}\right)^2} = \frac{1}{4\beta} \operatorname{sech}^2\left(\frac{x-\alpha}{2\beta}\right), \quad -\infty < x < \infty.$$

Because this function can be expressed in terms of the square of the hyperbolic secant function “sech”, it is sometimes referred to as the sech-square(d) distribution.

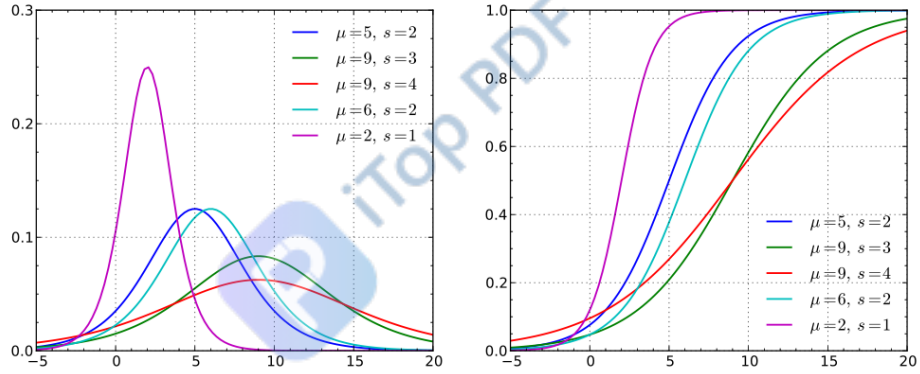


Figure 2.3: The *pdf* and *cdf* of Logistic distribution.

2.3.2 Cumulative Distribution Function

The logistic distribution receives its name from its cumulative distribution function, which is an instance of the family of logistic functions. The cumulative distribution function of the logistic distribution is also a scaled version of the hyperbolic tangent.

$$F(x; \alpha, \beta) = \frac{1}{1 + e^{-\frac{x-\alpha}{\beta}}} = \frac{1}{2} + \frac{1}{2} \tanh\left(\frac{x-\alpha}{2\beta}\right), \quad -\infty < x < \infty. \quad (2.10)$$

In this equation, x is the random variable, α is the mean, and β is a scale parameter proportional to the standard deviation.

2.3.3 Applications

The logistic distribution and the S -shaped pattern of its cumulative distribution function (the logistic function) and quantile function (the logit function) have been extensively used in many different areas. Some of which are

- (i) One of the most common applications is in logistic regression, which is used for modeling categorical dependent variables,
- (ii) In the theory of electron properties in semiconductors and metals,
- (iii) In hydrology the distribution of long duration river discharge and rainfall, and
- (iv) The United States Chess Federation and FIDE have switched its formula for calculating chess ratings from the normal distribution to the logistic distribution.

2.4 Erlang Distribution

The Erlang distribution is a two-parameter family of continuous probability distributions with support $x \in [0, \infty)$. The two parameters are:

- (i) a positive integer k , the “shape”, and
- (ii) a positive real number λ the “rate”. The “scale”, μ , the reciprocal of the rate, is sometimes used instead.

The Erlang distribution with shape parameter $k = 1$ simplifies to the exponential distribution. It is a special case of the gamma distribution. It is the distribution of a sum of k independent exponential variables with mean $\frac{1}{\lambda}$ each.

The Erlang distribution was developed by A. K. Erlang to examine the number of telephone calls which might be made at the same time to the operators of the switching

stations. This work on telephone traffic engineering has been expanded to consider waiting times in queueing systems in general. The distribution is also used in the field of stochastic processes.

2.4.1 Definition

The probability density function of the Erlang distribution is

$$f(x; k, \lambda) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{(k-1)!} \quad \text{for } x, \lambda \geq 0,$$

The parameter k is called the shape parameter, and the parameter λ is called the rate parameter.

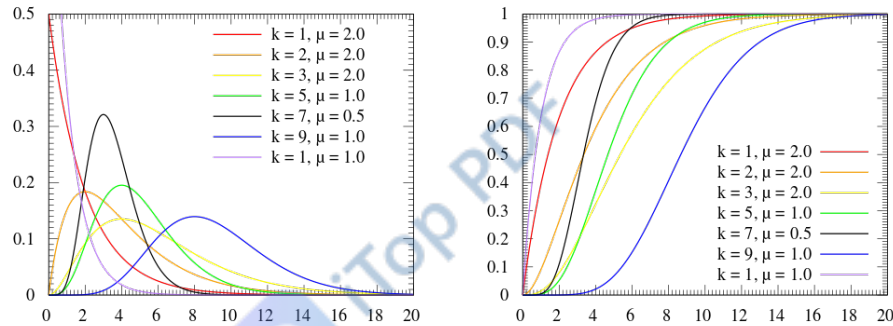


Figure 2.4: The *pdf* and *cdf* of Erlang distribution.

2.4.2 Applications

- (i) The age distribution of cancer incidence often follows the Erlang distribution, whereas the shape and scale parameters predict, respectively, the number of driver events and the time interval between them.
- (ii) More generally, the Erlang distribution has been suggested as good approximation of cell cycle time distribution, as result of multi-stage models.
- (iii) It has also been used in business economics for describing interpurchase times.

2.5 Maxwell Distribution

The Maxwell (Maxwell–Boltzmann) distribution is a particular probability distribution named after James Clerk Maxwell and Ludwig Boltzmann. It was first defined and used for describing particle speeds in idealized gases. Mathematically, the Maxwell–Boltzmann distribution is the chi distribution with three degrees of freedom.

2.5.1 Definition

A continuous random variable X is said to have a Maxwell distribution if its density function is defined as

$$f(x, \lambda) = \sqrt{\frac{2}{\pi}} \frac{x^2 e^{-\frac{x^2}{2\lambda^2}}}{\lambda^3}, \quad 0 \leq x < \infty,$$

where $\lambda > 0$ is the only parameter of the distribution.

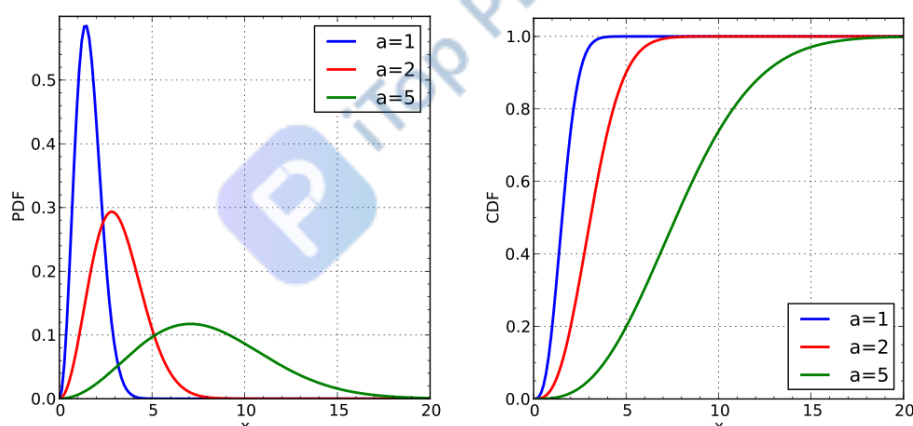


Figure 2.5: The *pdf* and *cdf* of Maxwell distribution.

2.5.2 Applications

The Maxwell distribution applies fundamentally to particle velocities in three dimensions. The kinetic theory of gases applies to the classical ideal gas, which is an idealization of real gases. However, rarefied gases at ordinary temperatures behave very nearly like an ideal gas and the Maxwell speed distribution is an excellent approximation for such gases. Ideal plasmas, which are ionized gases of sufficiently

low density, frequently also have particle distributions that are partially or entirely Maxwellian.

2.6 Weibull Distribution

The Weibull distribution, introduced by Waloddi Weibull (1951), is a popularly used statistical model in the area of reliability analysis.

2.6.1 Definition

The probability density function of a Weibull random variable is:

$$f(x; \lambda, k) = \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k} & x \geq 0, \\ 0 & x < 0, \end{cases}$$

where $k > 0$ is the shape parameter and $\lambda > 0$ is the scale parameter of the distribution. Its complementary cumulative distribution function is an exponential function.

The Weibull distribution is related to a number of other probability distributions; in particular, it interpolates between the exponential distribution ($k = 1$) and the Rayleigh distribution ($k = 2$ and $\lambda = 2\sigma$).

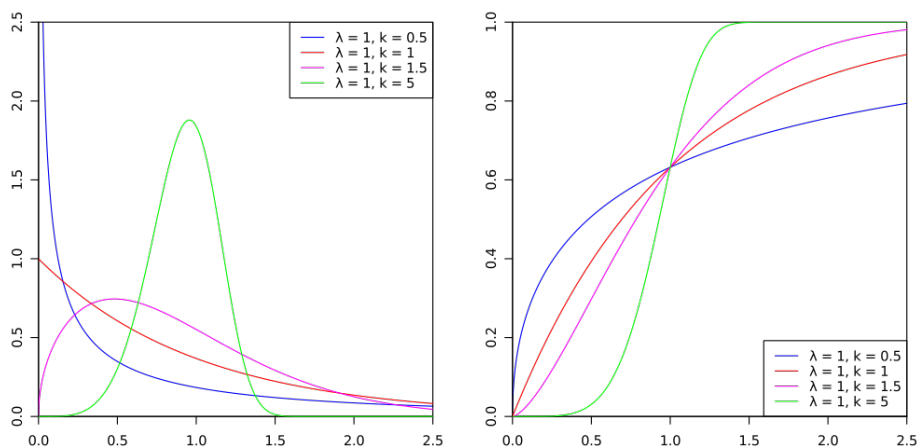


Figure 2.6: The *pdf* and *cdf* of Weibull distribution.

2.6.2 Applications

The Weibull distribution is used:

- (i) In survival analysis,
- (ii) In reliability engineering and failure analysis,
- (iii) In electrical engineering to represent overvoltage occurring in an electrical system,
- (iv) In industrial engineering to represent manufacturing and delivery times,
- (v) In extreme value theory,
- (vi) In weather forecasting and the wind power industry to describe wind speed distributions, as the natural distribution often matches the Weibull shape,
- (vii) In communications systems engineering,
- (viii) In information retrieval to model dwell times on web pages,
- (ix) In general insurance to model the size of reinsurance claims,
- (x) In forecasting technological change,
- (xi) In hydrology the Weibull distribution is applied to extreme events such as annual maximum one-day rainfalls and river discharges,
- (xii) In describing the size of particles generated by grinding, milling and crushing operations, the 2-Parameter Weibull distribution is used, and in these applications it is sometimes known as the Rosin-Rammler distribution.

2.7 Gumbel Distribution

The Gumbel distribution (Generalized Extreme Value distribution Type-I) is used to model the distribution of the maximum (or the minimum) of a number of samples of various distributions. This distribution might be used to represent the distribution

of the maximum level of a river in a particular year if there was a list of maximum values for the past ten years. It is useful in predicting the chance that an extreme earthquake, flood or other natural disaster will occur. The potential applicability of the Gumbel distribution to represent the distribution of maxima relates to extreme value theory, which indicates that it is likely to be useful if the distribution of the underlying sample data is of the normal or exponential type.

2.7.1 Definition

A continuous random variable X is said to have a Gumbel distribution if its cumulative distribution function is defined as

$$F(x; \mu, \beta) = e^{-e^{-(x-\mu)/\beta}}, \quad x \in \mathbb{R},$$

where $\mu \in \mathbb{R}$ and $\beta > 0$ are the two parameters of the distribution.

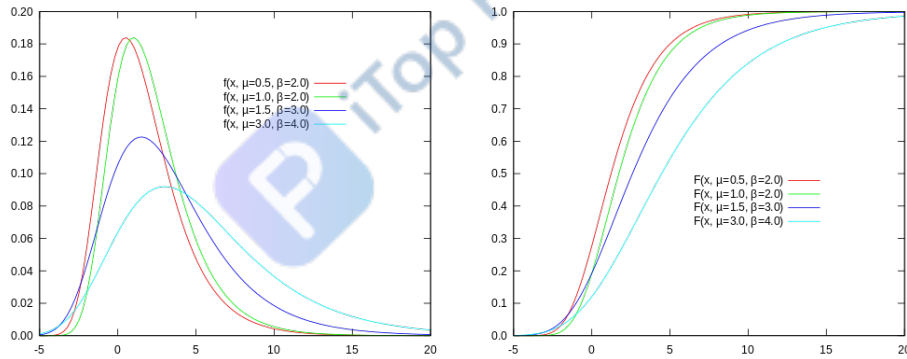


Figure 2.7: The *pdf* and *cdf* of Gumbel distribution.

2.7.2 Applications

- (i) Gumbel has shown that the maximum value (or last order statistic) in a sample of a random variable following an exponential distribution approaches the Gumbel distribution closer with increasing sample size,
- (ii) In hydrology, therefore, the Gumbel distribution is used to analyze such variables as monthly and annual maximum values of daily rainfall and river discharge volumes, and also to describe droughts,

- (iii) In number theory, the Gumbel distribution approximates the number of terms in a random partition of an integer,
- (iv) In machine learning, the Gumbel distribution is sometimes employed to generate samples from the categorical distribution.

2.8 Rayleigh Distribution

Rayleigh distribution is a continuous probability distribution for nonnegative-valued random variables. The Rayleigh distribution is a special case of the Weibull distribution with a scale parameter of 2. When a Rayleigh is set with a shape parameter (σ) of 1, it is equal to a chi square distribution with 2 degrees of freedom.

One example where the Rayleigh distribution naturally arises is when wind velocity is analyzed in two dimensions. Assuming that each component is uncorrelated, normally distributed with equal variance, and zero mean, then the overall wind speed (vector magnitude) will be characterized by a Rayleigh distribution. A second example of the distribution arises in the case of random complex numbers whose real and imaginary components are independently and identically distributed Gaussian with equal variance and zero mean. In that case, the absolute value of the complex number is Rayleigh-distributed.

2.8.1 Definition

The probability density function of Rayleigh distribution is given as

$$f_X(x) = \frac{2x}{\sigma^2} e^{-\frac{x^2}{\sigma^2}}, \quad x \in [0, \infty),$$

where σ is the scale parameter of the distribution.

The cumulative distribution function is

$$F(x) = 1 - e^{-x^2/(\sigma^2)}, \quad x \in [0, \infty).$$

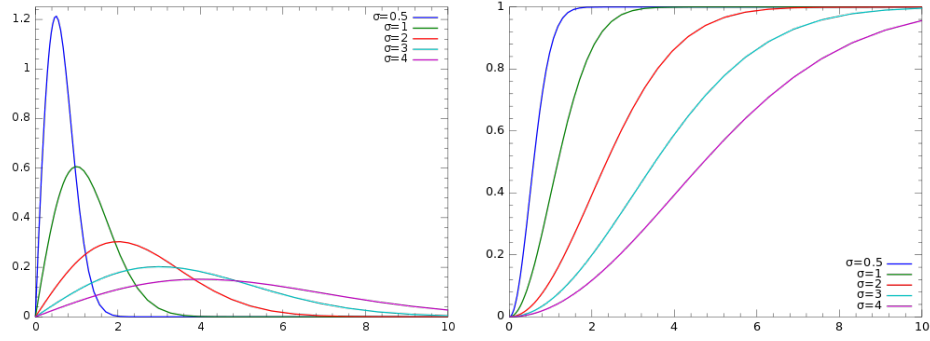


Figure 2.8: The pdf and cdf of Pareto distribution.

2.8.2 Moments

The r th raw moments can be easily obtain as

$$\begin{aligned}
 \mu'_r &= E(X^r) \\
 &= \int_0^\infty x^r \frac{2x}{\sigma^2} e^{\frac{-x^2}{\sigma^2}} dx \\
 &= \int_0^\infty \frac{2}{\sigma^2} x^{r+1} e^{\frac{-x^2}{\sigma^2}} dx \\
 &= \int_0^\infty x^r e^{\frac{-x^2}{\sigma^2}} \frac{2x}{\sigma^2} dx.
 \end{aligned} \tag{2.11}$$

Hence by using, $z = \frac{x^2}{\sigma^2} \Rightarrow \frac{2x}{\sigma^2} dx = dz$ and (2.11), we have

$$\begin{aligned}
 \mu'_r &= \int_0^\infty \left(z^{\frac{1}{2}}\sigma\right)^r e^{-z} dz \\
 &= \sigma^r \int_0^\infty z^{\frac{r}{2}} e^{-z} dz \\
 &= \sigma^r \int_0^\infty e^{-z} z^{\frac{r}{2}+1-1} dz \\
 &= \sigma^r \Gamma\left(\frac{r}{2} + 1\right).
 \end{aligned}$$

In particular

$$E(X) = \mu'_1 = 0.886\sigma \text{ and } V(X) = \mu'_2 - (\mu'_1)^2 = \sigma^2 - 0.785\sigma^2 = 0.215\sigma^2.$$

Assignment: Hence find skewness and kurtosis.

2.8.3 Mode

We have

$$\begin{aligned} f_X(x) &= \frac{2x}{\sigma^2} e^{-\frac{x^2}{\sigma^2}} \\ \Rightarrow \log f_X(x) &= \log \left(\frac{2x}{\sigma^2} \right) - \frac{x^2}{\sigma^2} \\ \Rightarrow \frac{d}{dx} \{ \log f_X(x) \} &= \frac{1}{x} - \frac{2x}{\sigma^2}. \end{aligned}$$

For finding mode, set $\frac{d}{dx} \{ \log f_X(x) \} = 0$ and obtain as

$$\begin{aligned} \frac{1}{x} - \frac{2x}{\sigma^2} &= 0 \\ \Rightarrow 2x^2 &= \sigma^2 \\ \therefore x &= \frac{1}{\sqrt{2}}\sigma. \end{aligned}$$

Hence, satisfies the equation $\frac{d^2}{dx^2} \{ \log f_X(x) \} < 0$ for $x = \frac{1}{\sqrt{2}}\sigma$.

So mode the distribution is $\frac{1}{\sqrt{2}}\sigma$.

2.8.4 Median

The cumulative distribution function is

$$F(x) = 1 - e^{-x^2/(\sigma^2)}, \quad x \in [0, \infty).$$

Let m be the median of this distribution, then

$$\begin{aligned} F(m) &= 1 - e^{-m^2/(\sigma^2)} \\ \Rightarrow \frac{1}{2} &= 1 - e^{-m^2/(\sigma^2)} \\ \Rightarrow \frac{1}{2} &= e^{-m^2/(\sigma^2)} \\ \Rightarrow 1 &= 2 e^{-m^2/(\sigma^2)} \\ \Rightarrow 0 &= \log 2 - m^2/(\sigma^2) \end{aligned}$$

$$\begin{aligned}\Rightarrow m^2 &= \sigma^2 \log 2 \\ \therefore m &= \sigma \sqrt{\log 2}.\end{aligned}$$

2.8.5 Applications

The Rayleigh distribution is a continuous probability distribution named after the English Lord Rayleigh. The distribution is widely used:

- (i) In communications theory, to model multiple paths of dense scattered signals reaching a receiver,
- (ii) In the physical sciences to model wind speed, wave heights and sound/light radiation,
- (iii) In engineering, to measure the lifetime of an object, where the lifetime depends on the object's age. For example: resistors, transformers, and capacitors in aircraft radar sets,
- (iv) In medical imaging science, to model noise variance in magnetic resonance imaging, and so on.

2.9 Pareto Distribution

The Pareto distribution developed by Vilfredo Pareto (1848-1923), state the allocation of wealth in a society. He observed that in most of the capitalist countries 80% wealth handled by its 20% population. He generalize this idea as Pareto's law or 80/20 principle, roughly followed the Pareto distribution.

2.9.1 Definition

If X is a random variable with a Pareto distribution, then probability density function is defined as follow

$$f_X(x) = \frac{\theta k^\theta}{x^{\theta+1}} ; \quad k \leq x < \infty,$$

where $\theta > 0$ and $k > 0$.

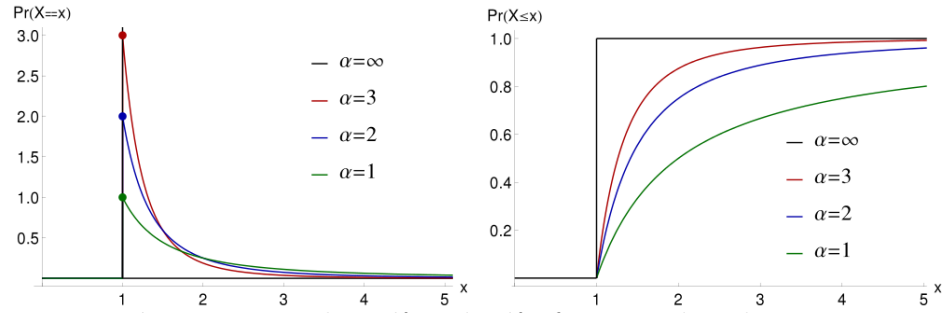


Figure 2.9: The pdf and cdf of Pareto distribution.

2.9.2 Mean and Variance of Pareto Distribution

If X follows a Pareto distribution, then mean can be obtain as

$$\begin{aligned}
 E(X) &= \int_k^{\infty} x \frac{\theta k^{\theta}}{x^{\theta+1}} dx \\
 &= \theta k^{\theta} \int_k^{\infty} x^{-\theta} dx \\
 &= \theta k^{\theta} \cdot \left. \frac{x^{-\theta+1}}{-\theta+1} \right|_k^{\infty} \\
 &= -\frac{\theta k^{\theta}}{\theta-1} \cdot [0 - k^{-\theta+1}], \quad [\text{if, } \theta > 1] \\
 &= \frac{\theta k}{\theta-1}, \quad [\text{if, } \theta > 1].
 \end{aligned}$$

Again variance can be determine as

$$V(X) = E(X^2) - \{E(X)\}^2. \quad (2.12)$$

Hence,

$$\begin{aligned}
 E(X^2) &= \int_k^{\infty} x^2 \frac{\theta k^{\theta}}{x^{\theta+1}} dx \\
 &= \theta k^{\theta} \int_k^{\infty} x^{-\theta+1} dx \\
 &= \theta k^{\theta} \cdot \left. \frac{x^{-\theta+2}}{-\theta+2} \right|_k^{\infty}
 \end{aligned}$$

$$\begin{aligned}
 &= -\frac{\theta k^\theta}{\theta - 2} \cdot [0 - k^{-\theta+2}], \quad [\text{if, } \theta > 2] \\
 &= \frac{\theta k^2}{\theta - 2}, \quad [\text{if, } \theta > 2], \quad [\text{if, } \theta > 2].
 \end{aligned}$$

Using equation (2.12), further proceed as

$$\begin{aligned}
 V(X) &= \frac{\theta k^2}{\theta - 2} - \left\{ \frac{\theta k}{\theta - 1} \right\}^2 \\
 &= \frac{\theta k^2(\theta - 1)^2 - \theta^2 k^2(\theta - 2)}{(\theta - 2)(\theta - 1)^2} \\
 &= \frac{\theta k^2(\theta^2 - 2\theta + 1) - \theta^3 k^2 + 2\theta^2 k^2}{(\theta - 2)(\theta - 1)^2} \\
 &= \frac{\theta^3 k^2 - 2\theta^2 k^2 + \theta k^2 - k^2 \theta^3 k^2 + 2\theta^2 k^2}{(\theta - 2)(\theta - 1)^2} \\
 &= \frac{\theta k^2}{(\theta - 2)(\theta - 1)^2}, \quad [\text{if, } \theta > 2].
 \end{aligned}$$

2.9.3 Moments

The r th raw moments can be easily obtain as

$$\begin{aligned}
 \mu'_r &= E(X^r) \\
 &= \int_k^\infty x^r \frac{\theta k^\theta}{x^{\theta+1}} dx \\
 &= \theta k^\theta \int_k^\infty x^{r-\theta-1} dx \\
 &= \theta k^\theta \cdot \left. \frac{x^{r-\theta}}{r-\theta} \right|_k^\infty \\
 &= \frac{\theta k^\theta}{r-\theta} \cdot [0 - k^{r-\theta}], \quad [\text{if, } \theta > r] \\
 &= \frac{\theta k^r}{\theta - r}, \quad [\text{if, } \theta > r].
 \end{aligned}$$

2.9.4 Geometric Mean

Let G be the geometric mean of the Pareto distribution, then

$$\begin{aligned}
 \log \left[\frac{G}{k} \right] &= E \left[\log \left(\frac{X}{k} \right) \right] \\
 &= \int_k^\infty \log \left(\frac{x}{k} \right) \frac{\theta k^\theta}{x^{\theta+1}} dx
 \end{aligned}$$

$$= \int_k^\infty \log\left(\frac{x}{k}\right) \frac{\theta}{k} \left(\frac{x}{k}\right)^{-(\theta+1)} dx. \quad (2.13)$$

Hence, let $\frac{x}{k} = z \Rightarrow dx = k dz$, then using (2.13), we have

$$\begin{aligned} \log\left[\frac{G}{k}\right] &= \int_1^\infty \log(z) \frac{\theta}{k} (z)^{-(\theta+1)} k dz \\ &= \theta \int_1^\infty \log(z) (z)^{-(\theta+1)} dz. \end{aligned} \quad (2.14)$$

Again let, $\log(z) = u \Rightarrow dz = z du = e^u dz$ and using (2.14), further written as

$$\begin{aligned} \log\left[\frac{G}{k}\right] &= \theta \int_0^\infty u e^{-(\theta+1)u} e^u du \\ \Rightarrow \log\left[\frac{G}{k}\right] &= \theta \int_0^\infty u e^{-\theta u} du \\ \Rightarrow \log\left[\frac{G}{k}\right] &= \theta \int_0^\infty e^{-\theta u} u^{2-1} du \\ \Rightarrow \log\left[\frac{G}{k}\right] &= \theta \frac{\Gamma(2)}{\theta^2} \\ \Rightarrow \log\left[\frac{G}{k}\right] &= \frac{1}{\theta} \\ \Rightarrow \frac{G}{k} &= e^{\frac{1}{\theta}} \\ \therefore G &= k e^{\frac{1}{\theta}}. \end{aligned}$$

For Pareto distribution geometric mean is more useful than arithmetic mean.

2.9.5 Applications

Vilfredo Pareto originally used this distribution to describe the allocation of wealth among individuals since it seemed to show rather well the way that a larger portion of the wealth of any society is owned by a smaller percentage of the people in that society. He also used it to describe distribution of income.

This distribution is not limited to describing wealth or income, but to many situations in which an equilibrium is found in the distribution of the “small” to the “large”. The following examples are sometimes seen as approximately Pareto-distributed:

- (i) The sizes of human settlements,
- (ii) Hard disk drive error rates,
- (iii) The values of oil reserves in oil fields,
- (iv) Amount of time a user on Steam will spend playing different games,
- (v) In hydrology the Pareto distribution is applied to extreme events such as annually maximum one-day rainfalls and river discharges,
- (vi) Occurrence of natural resources,
- (vii) Stock price fluctuations,
- (viii) Size of firms,
- (xi) Personal incomes and so on.

2.10 Inverse Gaussian Distribution

In probability theory, the inverse Gaussian distribution (also known as the Wald distribution) is a two-parameter family of continuous probability distributions with support on $(0, \infty)$. Its probability density function is given by

$$f(x) = \left[\frac{\lambda}{2\pi x^3} \right]^{\frac{1}{2}} \exp \left\{ -\frac{\lambda(x - \mu)^2}{2\mu^2 x} \right\},$$

for $x > 0$, where $\mu > 0$ is the mean and $\lambda > 0$ is the shape parameter.

As λ tends to infinity, the inverse Gaussian distribution becomes more like a normal (Gaussian) distribution. The inverse Gaussian distribution has several properties analogous to a Gaussian distribution. The name can be misleading: it is an “inverse” only in that, while the Gaussian describes a Brownian motion’s level at a fixed time, the inverse Gaussian describes the distribution of the time a Brownian motion with positive drift takes to reach a fixed positive level.

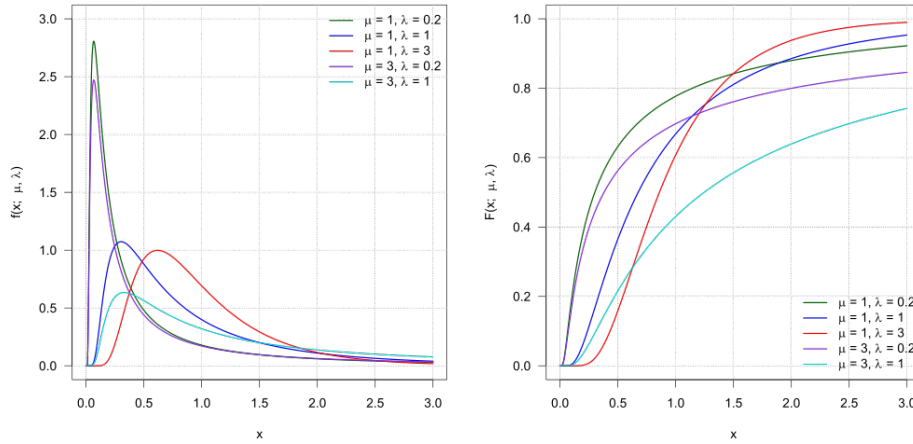


Figure 2.10: The *pdf* and *cdf* of Pareto distribution.

Its cumulant generating function (logarithm of the characteristic function) is the inverse of the cumulant generating function of a Gaussian random variable.

2.11 Triangular Distribution

The triangular distribution is a continuous probability distribution with lower limit a , upper limit b and mode c , where $a < b$ and $a \leq c \leq b$. The density function can be defined as follow

$$f(x) = \begin{cases} 0 & \text{for } x < a, \\ \frac{2(x-a)}{(b-a)(c-a)} & \text{for } a \leq x < c, \\ \frac{2}{b-a} & \text{for } x = c, \\ \frac{2(b-x)}{(b-a)(b-c)} & \text{for } c \leq x < b, \\ 0 & \text{for } x > b. \end{cases}$$

2.11.1 Applications

The triangular distribution is typically used as a subjective description of a population for which there is only limited sample data, and especially in cases where the relationship between variables is known but data is scarce (possibly because of the high cost of collection). It is based on a knowledge of the minimum and maximum

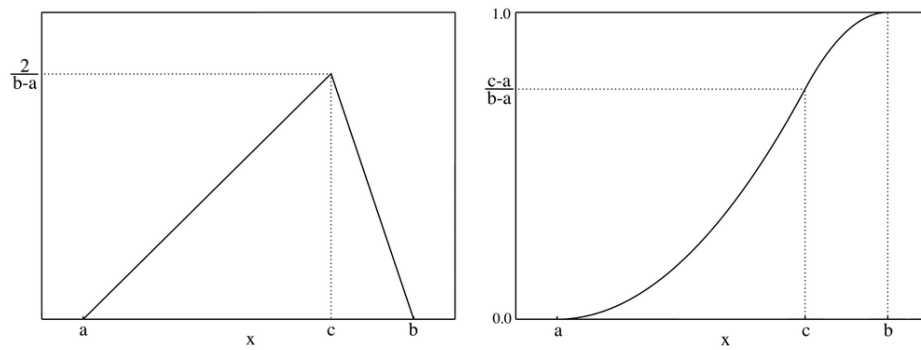


Figure 2.11: The *pdf* and *cdf* of Triangular distribution.

and an “inspired guess” as to the modal value. For these reasons, the triangle distribution has been called a “lack of knowledge” distribution.

Business simulations: The triangular distribution is therefore often used in business decision making, particularly in simulations. Generally, when not much is known about the distribution of an outcome (say, only its smallest and largest values), it is possible to use the uniform distribution. But if the most likely outcome is also known, then the outcome can be simulated by a triangular distribution. See for example under corporate finance.

Project management: The triangular distribution, along with the program evaluation and review technique (PERT) distribution, is also widely used in project management (as an input into PERT and hence critical path method (CPM)) to model events which take place within an interval defined by a minimum and maximum value.

Audio dithering: The symmetric triangular distribution is commonly used in audio dithering, where it is called triangular probability density function (TPDF).

2.12 Exponential family of Distributions

A single-parameter exponential family is a set of probability distributions whose probability density function (or probability mass function, for the case of a discrete dis-

tribution) can be expressed in the form

$$f_X(x \mid \theta) = h(x) \cdot \exp [\eta(\theta) \cdot T(x) - A(\theta)],$$

where $T(x)$, $h(x)$, $\eta(\theta)$, and $A(\theta)$ are known functions.

Example

Let $X \sim \text{Poisson}(\lambda)$ then,

$$\begin{aligned} p(x) &= \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots \\ &= \frac{1}{x!} \cdot \exp [\log(\lambda) \cdot x - \lambda], \end{aligned} \tag{2.15}$$

where $T(x) = x$, $h(x) = \frac{1}{x!}$, $\eta(\theta) = \log(\lambda)$, and $A(\theta) = \lambda$.

Example

Let $X \sim \text{Binomial}(n, p)$ then,

$$\begin{aligned} p(x) &= \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n \\ &= \binom{n}{x} \left(\frac{p}{1-p} \right)^x (1-p)^n \\ &= \binom{n}{x} \exp \left[\log \left(\frac{p}{1-p} \right) x + n \log(1-p) \right], \end{aligned}$$

where $T(x) = x$, $h(x) = \binom{n}{x}$, $\eta(\theta) = \log \left(\frac{p}{1-p} \right)$, and $A(\theta) = -n \log(1-p)$.

Exponential families include many of the most common distributions. Among many others, exponential families includes the following: normal, exponential, gamma, chi-squared, beta, Dirichlet, Bernoulli, categorical, Poisson, Wishart, inverse Wishart, geometric, binomial (with fixed number of trials), multinomial (with fixed number of trials), and negative binomial (with fixed number of failures).

2.13 Pearsonian system of Curves

The Pearson distribution is a family of continuous probability distributions. It was first published by Karl Pearson in 1895 and subsequently extended by him in 1901 and 1916 in a series of articles on biostatistics.

The Pearson family of distributions contains several distributions which have been developed from time to time. The first set of these distributions is proposed in 1895 and numbered as I to IV along with normal distribution as type V. Type V distributions were extended in 1901 and he also proposed type VI distributions. He further developed special cases with subtypes in 1916 and named them as VII through XII distributions. The Pearson family of distributions is defined below.

The probability density function $y = f(x)$ of Pearson family of distributions is defined as any valid solution to the following differential equation

$$\frac{f'(x)}{f(x)} + \frac{a + (x - \lambda)}{c_0 + c_1(x - \lambda) + c_2(x - \lambda)^2} = 0, \quad (2.16)$$

with:

$$\begin{aligned} c_0 &= \frac{4\beta_2 - 3\beta_1}{10\beta_2 - 12\beta_1 - 18}\mu_2, \\ c_1 &= a = \sqrt{\mu_2}\sqrt{\beta_1}\frac{\beta_2 + 3}{10\beta_2 - 12\beta_1 - 18}, \\ c_2 &= \frac{2\beta_2 - 3\beta_1 - 6}{10\beta_2 - 12\beta_1 - 18}. \end{aligned}$$

The Pearson types are characterized by second central moment μ_2 , squared skewness β_1 and traditional kurtosis β_2 .

The Criterion k

It has been observed from (2.16) that the parameter a obtains a stationary point and under certain conditions a mode of the distribution, since $f'(\lambda - a) = 0$ follows

directly from the differential equation. The Pearson family of distributions is obtained by solving the differential Equation (2.16) which can also be rewritten as

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= -\frac{(x-a)}{c_0 + c_1x + c_2x^2} \\ \Rightarrow \int \frac{1}{y} \frac{dy}{dx} dx &= -\int \frac{(x-a)}{c_0 + c_1x + c_2x^2} dx \\ \Rightarrow \log\left(\frac{y}{c}\right) &= -\int \frac{(x-a)}{c_0 + c_1x + c_2x^2} dx \\ \therefore y &= f(x) = ce^I,\end{aligned}$$

where c is the constant of integration and $I = -\int \frac{(x-a)}{c_0 + c_1x + c_2x^2} dx$. Hence, $y = f(x)$ depends on the roots of the quadratic equation $c_0 + c_1x + c_2x^2$, which can be further expressed as

$$\begin{aligned}c_0 + c_1x + c_2x^2 &= c_2 \left[x^2 + \frac{c_1}{c_2}x + \frac{c_0}{c_2} \right] \\ &= b_2 \left[\left(x + \frac{c_1}{2c_2} \right)^2 - \left(\frac{c_1^2}{4c_2^2} - \frac{c_0}{c_2} \right) \right] \\ &= b_2 \left[\left(x + \frac{c_1}{2c_2} \right)^2 - \frac{4c_0^2}{c_1^2} \frac{c_1^2}{4c_2c_0} \left(\frac{c_1^2}{4c_2c_0} - 1 \right) \right] \\ &= b_2 \left[\left(x + \frac{c_1}{2c_2} \right)^2 - \frac{4c_0^2}{c_1^2} k(k-1) \right],\end{aligned}$$

where $k = \frac{c_1^2}{4c_2c_0}$ and is used as a criterion for obtaining the form of density functions.

Table 2.1: Pearson Frequency Curves: Type I through Type VII

Criterion k	Corresponding Frequency Curve
$k < 0$	Type I
$k = 0$	Type II
$(\beta_1 = 0, \beta_2 < 3)$	Type III
$k \rightarrow \pm\infty$	Type IV
$0 < k < 1$	Type V
$k = 1$	Type VI
$k > 1$	Type VII
$k = 0$	Type VII
$(\beta_1 = 0, \beta_2 > 3)$	

Various choices of k leads to a specific type. These are described in Table 2.1. Basically the Pearson family of distributions is made up of seven distributions and remaining distributions are modifications of these.

2.14 Johnsons System of Curves

Johnson (1949) developed a flexible system of distributions, based on three families of transformations, that translate an observed, non-normal variate to one conforming to the standard normal distribution.

The exponential, logistic, and hyperbolic sine transformations are used to generate log-normal (S_L), unbounded (S_U), and bounded (S_B) distributions, respectively. The coefficients defining a Johnson distribution consist of two shape (γ, δ), a location (ξ), and a scale (λ) parameter.

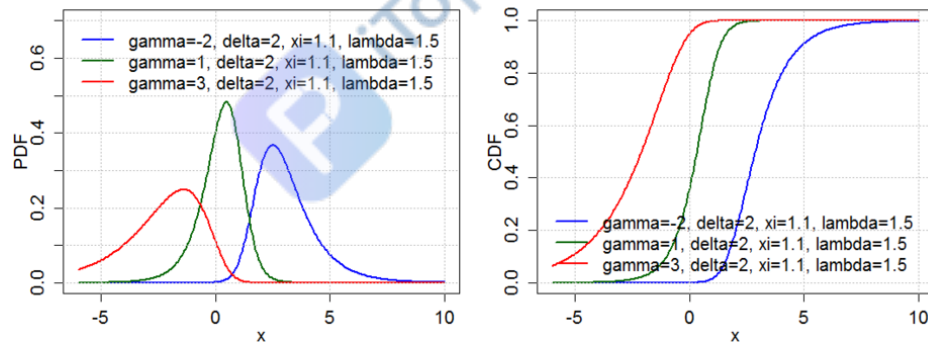


Figure 2.12: The *pdf* and *cdf* of Johnson's S_U distribution.

The Johnson's S_U -distribution is a four-parameter family of probability distributions first investigated by N. L. Johnson in 1949, can be written as

$$f(x) = \frac{\delta}{\lambda\sqrt{2\pi}} \frac{1}{\sqrt{1 + \left(\frac{x-\xi}{\lambda}\right)^2}} e^{-\frac{1}{2}\left(\gamma + \delta \sinh^{-1}\left(\frac{x-\xi}{\lambda}\right)\right)^2}, \quad x \in \mathbb{R}.$$

Johnson proposed it as a transformation of the normal distribution:

$$z = \gamma + \delta \sinh^{-1}\left(\frac{x - \xi}{\lambda}\right),$$

where $z \sim N(0, 1)$.

Johnson's S_B -distribution is defined as

$$f(x) = \frac{\delta}{\lambda\sqrt{2\pi}} \frac{1}{\sqrt{1 + \left(\frac{x-\xi}{\lambda}\right)^2}} e^{-\frac{1}{2}\left(\gamma + \delta \sinh^{-1}\left(\frac{x-\xi}{\lambda}\right)\right)^2}, \quad x \in \mathbb{R}.$$

and proposes the transformation :

$$z = \gamma + \delta \sinh^{-1} \left(\frac{x - \xi}{\lambda} \right),$$

where $z \sim N(0, 1)$.

Johnson's S_U -distribution has been used successfully to model asset returns for portfolio management.

2.15 Truncated Distributions

Truncated distribution is a conditional distribution that results from restricting the domain of some other probability distribution. Truncated distributions arise in practical statistics in cases where the ability to record, or even to know about, occurrences is limited to values which lie above or below a given threshold or within a specified range. *For example*, if the dates of birth of children in a school are examined, these would typically be subject to truncation relative to those of all children in the area given that the school accepts only children in a given age range on a specific date.

2.15.1 Definition

The following discussion is in terms of a random variable having a continuous distribution although the same ideas apply to discrete distributions. Similarly, the discussion assumes that truncation is to a semi-open interval $y \in (a, b]$ but other possibilities can be handled straightforwardly.

Suppose we have a random variable, X that is distributed according to some prob-

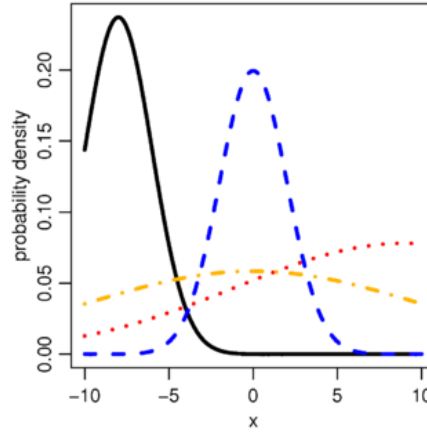


Figure 2.13: Truncated Distributions.

ability density function, $f(x)$, with cumulative distribution function $F(x)$ both of which have infinite support. Suppose we wish to know the probability density of the random variable after restricting the support to be between two constants so that the support, $y = (a, b]$. That is to say, suppose we wish to know how X is distributed given $a < X \leq b$.

$$f(x|a < X \leq b) = \frac{g(x)}{F(b) - F(a)} = Tr(x),$$

where $g(x) = f(x)$ for all $a < x \leq b$ and $g(x) = 0$ everywhere else. Notice that $Tr(x)$ has the same support as $g(x)$. Notice that in fact $f(x|a < X \leq b)$ is a density:

$$\int_a^b f(x|a < X \leq b)dx = \frac{1}{F(b) - F(a)} \int_a^b g(x)dx = 1.$$

Truncated distributions need not have parts removed from the top and bottom. A truncated distribution where just the bottom of the distribution has been removed is as follows:

$$f(x|X > y) = \frac{g(x)}{1 - F(y)},$$

where $g(x) = f(x)$ for all $y < x$ and $g(x) = 0$ everywhere else, and $F(x)$ is the cumulative distribution function. A truncated distribution where the top of the distribution

has been removed is as follows:

$$f(x|X \leq y) = \frac{g(x)}{F(y)},$$

where $g(x) = f(x)$ for all $x \leq y$ and $g(x) = 0$ everywhere else, and $F(x)$ is the cumulative distribution function.

2.15.2 Truncated Binomial Distribution

2.15.2.1 Binomial Distribution Truncated at $x = 0$

A discrete random variable X having the probability function

$$p(x) = \frac{1}{1 - q^n} \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 1, 2, \dots, n,$$

is called binomial random variable truncated at $x = 0$. The distribution is called binomial distribution truncated at $x = 0$. This distribution is sometimes called the positive binomial distribution. The mean of the distribution is

$$\begin{aligned} \mu &= \frac{1}{1 - q^n} \sum_{x=1}^n x \binom{n}{x} p^x (1 - p)^{n-x} \\ &= \frac{np}{1 - q^n} \sum_{x=1}^n x \frac{(n-1)!}{(n-x)!x!} p^{x-1} (1-p)^{(n-1)-(x-1)} \\ &= \frac{np}{1 - q^n} \sum_{x=1}^n \frac{(n-1)!}{((n-1)-(x-1))!(x-1)!} p^{x-1} (1-p)^{(n-1)-(x-1)} \\ &= \frac{np}{1 - q^n} \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} (1-p)^{(n-1)-(x-1)} \\ &= \frac{np}{1 - q^n} \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} p^{\ell} (1-p)^{(n-1)-\ell} && \text{with } \ell := x - 1 \\ &= \frac{np}{1 - q^n} \sum_{\ell=0}^m \binom{m}{\ell} p^{\ell} (1-p)^{m-\ell} && \text{with } m := n - 1 \\ &= \frac{np}{1 - q^n} (p + (1-p))^m \\ &= \frac{np}{1 - q^n}. \end{aligned}$$

and

$$\begin{aligned}
 \sigma^2 &= E[X^2] - \{E[X]\}^2 \\
 &= E[X(X-1) + X] - \left\{ \frac{np}{1-q^n} \right\}^2 \\
 &= E[X(X-1)] + E[X] - \frac{n^2 p^2}{(1-q^n)^2} \\
 &\vdots
 \end{aligned}$$

Assignment

2.15.2.2 Binomial Distribution Truncated at $x = n$

A discrete random variable X having the probability function

$$p(x) = \frac{1}{1-p^n} \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, (n-1),$$

is called binomial random variable truncated at $x = n$. The distribution is called binomial distribution truncated at $x = n$.

2.15.3 Truncated Poisson Distribution

A discrete random variable X is said to have a truncated Poisson distribution at $x = 0$ if its density function is given by

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x! (1 - e^{-\lambda})}, \quad x = 1, 2, \dots, \infty.$$

Mean of the distribution is

$$\begin{aligned}
 \mu &= \frac{1}{1 - e^{-\lambda}} \sum_{x=1}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} \\
 &= \frac{e^{-\lambda} \lambda}{1 - e^{-\lambda}} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \\
 &= \frac{e^{-\lambda} \lambda}{1 - e^{-\lambda}} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} \quad [x-1 = y] \\
 &= \frac{e^{-\lambda} \lambda}{1 - e^{-\lambda}} e^{\lambda}
 \end{aligned}$$

$$= \frac{\lambda}{1 - e^{-\lambda}}.$$

and the variance is

$$\begin{aligned} \sigma^2 &= E[X^2] - \{E[X]\}^2 \\ &= E[X(X-1) + X] - \left\{ \frac{\lambda}{1 - e^{-\lambda}} \right\}^2 \\ &= E[X(X-1)] + E[X] - \frac{\lambda^2}{(1 - e^{-\lambda})^2} \\ &= \frac{\lambda^2}{1 - e^{-\lambda}} + \frac{\lambda}{1 - e^{-\lambda}} - \frac{\lambda^2}{(1 - e^{-\lambda})^2} \\ &= \frac{\lambda}{1 - e^{-\lambda}} \left[\lambda + 1 - \frac{\lambda}{1 - e^{-\lambda}} \right] \\ &= \frac{\lambda}{1 - e^{-\lambda}} \left[\frac{\lambda - \lambda e^{-\lambda} + 1 - e^{-\lambda} - \lambda}{1 - e^{-\lambda}} \right] \\ &= \frac{\lambda}{(1 - e^{-\lambda})^2} [1 - \lambda e^{-\lambda} - e^{-\lambda}]. \end{aligned}$$

2.15.4 Truncated Normal Distribution

The truncated normal distribution is the probability distribution derived from that of a normally distributed random variable by bounding the random variable from either below or above (or both). The truncated normal distribution has wide applications in statistics and econometrics.

Suppose X has a normal distribution with mean μ and variance σ^2 and lies within the interval (a, b) , with $-\infty \leq a < b \leq \infty$. Then X conditional on $a < X < b$ has a truncated normal distribution. Its probability density function, $f(x)$, for $a \leq x \leq b$, is given by

$$f(x; \mu, \sigma, a, b) = \frac{\phi\left(\frac{x-\mu}{\sigma}\right)}{\sigma \left(\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right) \right)}$$

and by $f = 0$ otherwise. Here,

$$\phi(\xi) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\xi^2\right),$$

is the probability density function of the standard normal distribution and $\Phi(\cdot)$ is its cumulative distribution function

$$\Phi(x) = \frac{1}{2} \left(1 + \operatorname{erf}(x/\sqrt{2}) \right).$$

By definition, if $b = \infty$, then $\Phi\left(\frac{b-\mu}{\sigma}\right) = 1$, and similarly, if $a = -\infty$, then $\Phi\left(\frac{a-\mu}{\sigma}\right) = 0$.

2.16 Compound and Mixture Distributions

Compound and mixture distributions are described by the following two subsections.

2.16.1 Compound Distribution:

A compound probability distribution is the probability distribution that results from assuming that a random variable X is distributed according to some parametrized distribution F with an unknown parameter θ that is again distributed according to some other distribution G . The resulting distribution H is said to be the distribution that results from compounding F with G . The parameter's distribution G is also called the mixing distribution or latent distribution. Technically, the unconditional distribution H results from marginalizing over G , i.e., from integrating out the unknown parameter(s) θ . Its probability density function is given by:

$$p_H(x) = \int p_F(x|\theta) p_G(\theta) d\theta.$$

From the above formula, one can see that a compound distribution essentially is a special case of a marginal distribution: The joint distribution of x and θ is given by $p(x, \theta) = p(x|\theta)p(\theta)$, and the compound results as its marginal distribution: $p(x) = \int p(x, \theta) d\theta$. If the domain of θ is discrete, then the distribution is again a special case of a mixture distribution.

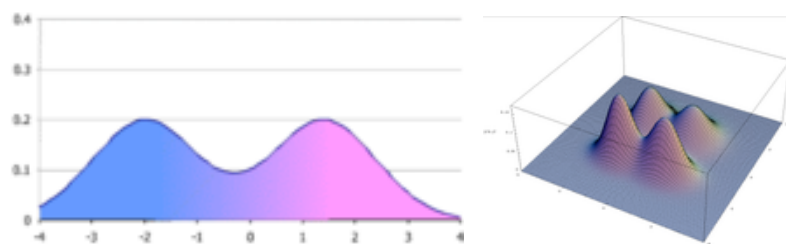


Figure 2.14: **Left:** Univariate mixture distribution, showing bimodal distribution; **Right:** Multivariate mixture distribution, showing four modes .

2.16.2 Mixture Distribution:

As an example, the sum of two jointly normally distributed random variables, each with different means, will still have a normal distribution. On the other hand, a mixture density created as a mixture of two normal distributions with different means will have two peaks provided that the two means are far enough apart, showing that this distribution is radically different from a normal distribution.

Mixture distributions arise in many contexts in the literature and arise naturally where a statistical population contains two or more subpopulations. They are also sometimes used as a means of representing non-normal distributions. Data analysis concerning statistical models involving mixture distributions is discussed under the title of mixture models, while the present article concentrates on the finite and countable mixtures.

Finite and Countable Mixture: Given a finite set of probability density functions $p_1(x), p_2(x), \dots, p_n(x)$, or corresponding cumulative distribution functions $P_1(x), P_2(x), \dots, P_n(x)$ and weights $w_1(x), w_2(x), \dots, w_n(x)$ such that $w_i \geq 0$ and $\sum w_i = 1$, the mixture distribution can be represented by writing either the density, f , or the distribution function, F , as a sum (which in both cases is a convex combination):

$$F(x) = \sum_{i=1}^n w_i P_i(x),$$

$$f(x) = \sum_{i=1}^n w_i p_i(x).$$

This type of mixture, being a finite sum, is called a finite mixture, and in applications, an unqualified reference to a mixture density usually means a finite mixture. The case of a countably infinite set of components is covered formally by allowing $n = \infty$.



Chapter 3

Variate Transformation

3.1 Variate Transformation

Let T be any statistic which is asymptotically normally distributed with mean θ and variance $\Psi(\theta)$, where $\Psi(\theta)$ is some function of the parameter θ , i.e., $T \sim N\{\theta, \Psi(\theta)\}$. Let us transform T by a function g as $g(T)$ where g is a function which possesses first order derivative which is continuous and $g'(\theta) \neq 0$, where $(')$ denotes differentiation with respect to the parameter θ . Then $g(T)$ is normally distributed about mean $g(\theta)$ and variance $[g'(\theta)]^2 \cdot \Psi(\theta)$, i.e.,

$$g(T) = N \left[g(\theta), [g'(\theta)]^2 \cdot \Psi(\theta) \right], \quad (3.1)$$

asymptotically, provided $g'(\theta) \neq 0$ is continuous in the neighbourhood of θ .

In general $Var(T) = \Psi(\theta)$ will be dependent on the parameter θ . We are interested in obtaining a function g such that the asymptotic variance of the transformed statistic $g(T)$ is independent of θ , i.e., it should be constant. In other words, we want g such that $Var[g(T)] = [g'(\theta)]^2 \cdot \Psi(\theta) = \text{Constant} = c^2$, (say) i.e.,

$$\left| g'(\theta) \right| = \frac{c}{\sqrt{\Psi(\theta)}},$$

Integrating both sides with respect to θ , we get

$$g(\theta) = \int \frac{c}{\sqrt{\Psi(\theta)}} d\theta. \quad (3.2)$$

3.2 Types of Variate Transformation

The commonly used transformations are:

- (i) Square root transformation,
- (ii) Sine inverse ($\sin^{-1}$) transformation,
- (iii) Logarithmic transformation, and
- (iv) Fisher's Z -Transformation.

3.2.1 Square Root Transformation

Square root transformation is a transformation for the Poisson variate. If a variable X follows Poisson distribution with parameter λ (assumed to be large), then we know that asymptotic distribution of X is normal (as $\lambda \rightarrow \infty$) with $E(X) = \lambda$ and $Var(X) = \lambda = \Psi(\lambda)$, in the above notations. Then (3.2) gives the function

$$g(\lambda) = \int \frac{c}{\sqrt{\lambda}} d\lambda = 2c\sqrt{\lambda}.$$

Now we select c in such a way that $2c = 1$, i.e., $c = \frac{1}{2}$ so that $g(\lambda) = \sqrt{\lambda}$.

Hence the transformed variable is $g(X) = \sqrt{X}$. Using (3.1), the transformed variable $\sqrt{X}$ has the mean

$$g(\lambda) = \sqrt{\lambda}$$

and variance

$$Var(\sqrt{X}) = [g'(\lambda)]^2 \cdot \Psi(\lambda)$$

$$\begin{aligned} &= \left[\frac{1}{2\sqrt{\lambda}} \right]^2 \cdot \lambda \\ &= \frac{1}{4}. \end{aligned}$$

Or alternatively

$$\text{Var}(\sqrt{X}) = c^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}.$$

Hence $\sqrt{X} \sim N\left(\sqrt{\lambda}, \frac{1}{4}\right)$, asymptotically.

3.2.2 Sine Inverse ($\sin^{-1}$) Transformation

Sine-inverse is the transformation for stabilizing the variance of a binomial variate. If p is the observed proportion of successes in a series of n independent trials with constant probability P of success for each trial then we know that the asymptotic distribution of p is asymptotically normal (as $n \rightarrow \infty$) with mean

$$E(p) = P,$$

and

$$\text{Var}(p) = \frac{PQ}{n}, \quad Q = 1 - P.$$

Hence $p \sim N\left(P, \frac{PQ}{n}\right)$, as $n \rightarrow \infty$.

In the usual notations

$$\Psi(p) = \frac{PQ}{n} = \frac{P(1-P)}{n}.$$

Using (3.2), the transforming function is

$$g(P) = \int \frac{c}{\sqrt{\Psi(P)}} dP$$

$$\begin{aligned}
 &= c\sqrt{n} \int \frac{dP}{\sqrt{P(1-P)}} \\
 &= 2c\sqrt{n} \sin^{-1}(\sqrt{P}) \\
 &\quad \left[\because \frac{d}{dP} \sin^{-1}(\sqrt{P}) = \frac{1}{\sqrt{1-P}} \cdot \frac{1}{2\sqrt{P}} \right]
 \end{aligned}$$

Choosing the constant c such that

$$2c\sqrt{n} = 1 \quad \Rightarrow c = \frac{1}{2\sqrt{n}},$$

we get

$$g(P) = \sin^{-1}(\sqrt{P}).$$

Hence the transformed statistic is

$$g(p) = \sin^{-1}\sqrt{p}.$$

Using (3.1), $g(p)$ has mean $\sin^{-1}\sqrt{P}$ and

$$\begin{aligned}
 \text{Var}(\sin^{-1}\sqrt{p}) &= [g'(P)]^2 \cdot \Psi(P) \\
 &= \left[\frac{1}{\sqrt{1-P}} \cdot \frac{1}{2\sqrt{P}} \right]^2 \cdot \frac{P(1-P)}{n} \\
 &= \frac{1}{4n}.
 \end{aligned}$$

Or alternatively $\text{Var}(\sin^{-1}\sqrt{p}) = c^2 = \left(\frac{1}{2\sqrt{n}}\right)^2 = \frac{1}{4n}$.

Hence $\sin^{-1}\sqrt{p} \sim N\left(\sin^{-1}\sqrt{P}, \frac{1}{4n}\right)$, asymptotically.

3.2.3 Logarithmic Transformation

Log transformation is the transformation for stabilizing the variance of the distribution of sample variance. If s^2 is the sample variance in a sample of size n from normal population with variance σ^2 , then the sampling distribution of s^2 is asymptotically

normal (as $n \rightarrow \infty$) with mean

$$E(s^2) = \sigma^2,$$

and

$$\text{Var}(s^2) = \frac{2\sigma^4}{n}.$$

In the usual notations we have

$$\Psi(\sigma^2) = \frac{2\sigma^4}{n}.$$

Using (3.2), the transformation function is

$$\begin{aligned} g(\sigma^2) &= \int \frac{c\sqrt{n}}{\sqrt{2}\sigma^2} d\sigma^2 \\ &= \frac{c\sqrt{n}}{\sqrt{2}} \log_e \sigma^2. \end{aligned}$$

We select c in such a way that

$$\frac{c\sqrt{n}}{\sqrt{2}} = 1 \quad \Rightarrow \quad c = \frac{\sqrt{2}}{\sqrt{n}},$$

so that

$$g(\sigma^2) = \log_e \sigma^2.$$

Hence the transformation for the statistic s^2 is

$$g(s^2) = \log_e s^2,$$

and using (3.1), the transformed statistic is normally distributed with mean

$$g(\sigma^2) = \log_e \sigma^2,$$

and variance

$$\begin{aligned} \text{Var} [g(s^2)] &= [g'(\sigma^2)]^2 \cdot \Psi(\sigma^2) \\ &= \left(\frac{1}{\sigma^2}\right)^2 \cdot \frac{2\sigma^4}{n} \\ &= \frac{2}{n}. \end{aligned}$$

Or alternatively

$$\text{Var} [g(s^2)] = c^2 = \left(\frac{\sqrt{2}}{\sqrt{n}}\right)^2 = \frac{2}{n}.$$

Hence $\log_e s^2 \sim N(\log_e \sigma^2, \frac{2}{n})$, for large n .

3.2.4 Fisher's Z -Transformation

This transformation is suggested for stabilizing the variance of the distribution of correlation coefficient. If r is the sample correlation coefficient in sampling from a correlated bivariate normal population with correlation coefficient ρ then the asymptotic distribution of r , as $n \rightarrow \infty$ is normal with mean

$$E(r) = \rho,$$

and

$$\text{Var}(r) = \frac{(1 - \rho^2)^2}{n}.$$

In the usual notations we have

$$\Psi(\rho) = \frac{(1 - \rho^2)^2}{n}.$$

Using (3.2), the transformation function is

$$g(\rho) = \int \frac{c\sqrt{n}}{1 - \rho^2} d\rho$$

$$= c\sqrt{n} \cdot \frac{1}{2} \log_e \left(\frac{1+\rho}{1-\rho} \right).$$

We select c in such a way that

$$c\sqrt{n} = 1 \quad \Rightarrow \quad c = \frac{1}{\sqrt{n}},$$

so that

$$g(\rho) = \frac{1}{2} \cdot \log_e \left(\frac{1+\rho}{1-\rho} \right).$$

Hence the transformation for the statistic is

$$g(r) = \frac{1}{2} \cdot \log_e \left(\frac{1+r}{1-r} \right),$$

which is denoted by Z , and using (3.1), the transformed statistic is normally distributed with mean

$$g(\rho) = \frac{1}{2} \cdot \log_e \left(\frac{1+\rho}{1-\rho} \right),$$

and variance

$$\begin{aligned} \text{Var}[g(r)] &= [g'(\rho)]^2 \cdot \Psi(\rho) \\ &= \left(\frac{1}{1-\rho^2} \right)^2 \cdot \frac{(1-\rho^2)^2}{n} \\ &= \frac{1}{n}, \text{ for large } n. \end{aligned}$$

Or alternatively

$$\text{Var}[g(r)] = c^2 = \left(\frac{1}{\sqrt{n}} \right)^2 = \frac{1}{n}.$$

Hence $Z = \frac{1}{2} \cdot \log_e \left(\frac{1+r}{1-r} \right) \sim N \left(\frac{1}{2} \cdot \log_e \left(\frac{1+\rho}{1-\rho} \right), \frac{1}{n} \right)$, for large n .

3.3 Applications

The transformation of a statistic T to a function $g(T)$, the distribution of $g(T)$ is approximately normal and its asymptotic variance is independent of the population parameter θ . Hence the use of statistic $g(T)$ gives better results and confidence intervals than the original statistic T .



Chapter 4

Sequences

4.1 Real Number Chart and Line

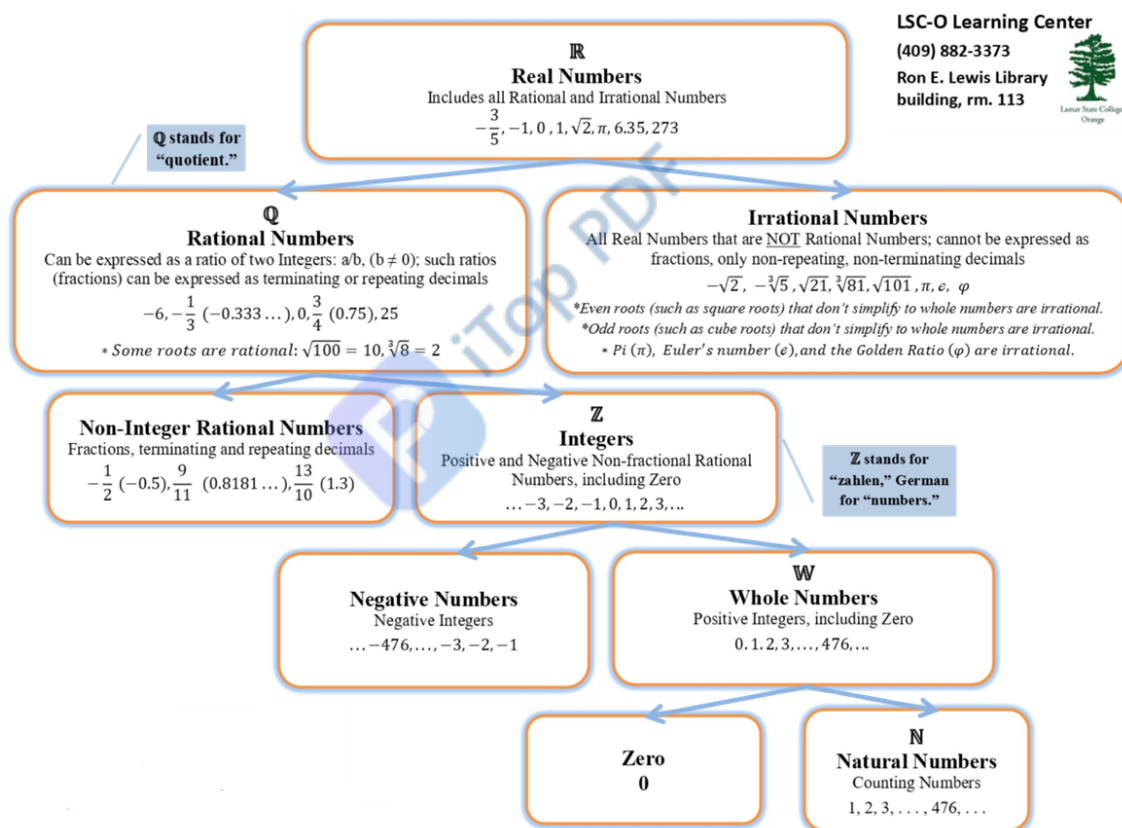


Figure 4.1: The Real Number Chart.

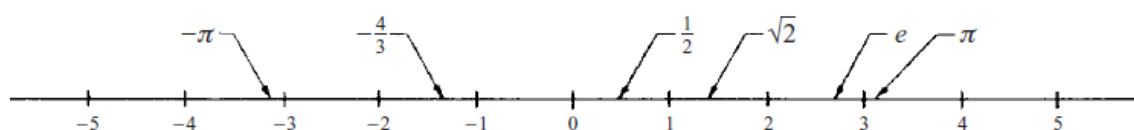


Figure 4.2: The Real Number Line.

4.2 Sequence

A sequence is a set of numbers $u_1, u_2, \dots$ in a definite order of arrangement (i.e., correspondence with the natural numbers) and formed according to a definite rule. Each number in the sequence is termed as u_n is called the n th term. The sequence is called finite or infinite according as there are or are not a finite number of terms. The sequence $u_1, u_2, \dots$ is also designated briefly by $\{u_n\}$.

Examples:

1. The set of numbers $2, 7, 12, 17, \dots, 32$ is a finite sequence; the n th term of the sequence is given by

$$\begin{aligned}u_n &= 2 + 5(n - 1) \\&= 5n - 3, \quad n = 1, 2, \dots, 7.\end{aligned}$$

2. The set of numbers $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$ is an infinite sequence; the n th term of the sequence is given as

$$u_n = \frac{1}{2n - 1}, \quad n = 1, 2, 3, \dots$$

Unless otherwise specified, we shall consider infinite sequence only.

4.3 limit of a Sequence

A number l is called the limit of an infinite sequence $u_1, u_2, \dots$ if for any positive number ϵ we can find a positive number N depending on ϵ such that $|u_n - l| < \epsilon$ for all integers $n > N$. In such case we write $\lim_{n \rightarrow \infty} u_n = l$.

Example:

If $u_n = 3 + \frac{1}{n}$, the sequence is $4, \frac{7}{2}, \frac{10}{2}, \dots$ and we can show that $\lim_{n \rightarrow \infty} u_n = 3$.

If the limit of a sequence exists, the sequence is called *convergent sequence*; otherwise, it is called *divergent sequence*. A sequence can converge to only one limit, i.e., if a limit exists, it is unique.

Example: Prove that $\lim_{n \rightarrow \infty} \frac{c}{n^p} = 0$ where $c \neq 0$ and $p > 0$ are constants.

Solution: We must show that for any $\epsilon > 0$ there is a number N such that $|\frac{c}{n^p} - 0| < \epsilon$ for all $n > N$.

Now $|\frac{c}{n^p}| < \epsilon$ when $\frac{|c|}{n^p} < \epsilon$, i.e., $n^p > \frac{|c|}{\epsilon}$ or $n > \left(\frac{|c|}{\epsilon}\right)^{\frac{1}{p}}$.

Choosing $N = \left(\frac{|c|}{\epsilon}\right)^{\frac{1}{p}}$ (depending on ϵ), we see that $|\frac{c}{n^p}| < \epsilon$ for all $n > N$, proving that $\lim_{n \rightarrow \infty} \frac{c}{n^p} = 0$.

4.4 Theorems on Limit of Sequences

Theorem 4.4.1. If $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$, then

1. $\lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n = A + B$
2. $\lim_{n \rightarrow \infty} (a_n - b_n) = \lim_{n \rightarrow \infty} a_n - \lim_{n \rightarrow \infty} b_n = A - B$
3. $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = \left(\lim_{n \rightarrow \infty} a_n\right) \left(\lim_{n \rightarrow \infty} b_n\right) = AB$
4. $\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n}\right) = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n} = \frac{A}{B}$, if $\lim_{n \rightarrow \infty} b_n = B \neq 0$
 - (i) If $B = 0$ and $A \neq 0$, $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ does not exist.
 - (ii) If $B = 0$ and $A = 0$, $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ may or may not exist.
5. $\lim_{n \rightarrow \infty} a_n^p = \left(\lim_{n \rightarrow \infty} a_n\right)^p = A^p$, for $p =$ any real number if A^p exists.
6. $\lim_{n \rightarrow \infty} p^{a_n} = p^{\lim_{n \rightarrow \infty} a_n} = p^A$, for $p =$ any real number if p^A exists.

Theorem 4.4.2. *If $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$, then*

$$\lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n = A + B$$

Proof. We must show that for any $\epsilon > 0$, we can find $N > 0$ such that $|(a_n + b_n) - (A + B)| < \epsilon$ for all $n > N$.

Now we can obtain as

$$\begin{aligned} |(a_n + b_n) - (A + B)| &= |(a_n - A) + (b_n - B)| \\ &\leq |a_n - A| + |b_n - B| \end{aligned} \quad (4.1)$$

By hypothesis, given $\epsilon > 0$ we can find N_1 and N_2 such that

$$|a_n - A| < \frac{\epsilon}{2} \quad \text{for all } n > N_1 \quad (4.2)$$

$$|b_n - B| < \frac{\epsilon}{2} \quad \text{for all } n > N_2 \quad (4.3)$$

Then from (4.1), (4.2) and (4.3) we have

$$|(a_n + b_n) - (A + B)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \quad \text{for all } n > N,$$

where N is chosen as the larger of N_1 and N_2 . Thus the required result follows. $\square$

Theorem 4.4.3. *Prove that if $\lim_{n \rightarrow \infty} u_n$ exists, it must be unique.*

Proof. We must show that if $\lim_{n \rightarrow \infty} u_n = l_1$ and $\lim_{n \rightarrow \infty} u_n = l_2$, then $l_1 = l_2$.

By hypothesis, given $\epsilon > 0$ we can find N such that

$$|u_n - l_1| < \frac{\epsilon}{2} \quad \text{when } n > N, \quad |u_n - l_2| < \frac{\epsilon}{2} \quad \text{when } n > N.$$

Then

$$\begin{aligned} |l_1 - l_2| &= |l_1 - u_n + u_n - l_2| \\ &\leq |l_1 - u_n| + |u_n - l_2| \end{aligned}$$

$$< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

i.e., $|l_1 - l_2|$ is less than any positive ϵ (however small) and so must be zero. Hence we can write as $l_1 = l_2$. $\square$

Theorem 4.4.4. *Prove that a convergent sequence is bounded.*

Proof. Given $\lim_{n \rightarrow \infty} a_n = A$, we must show that there exists a positive number P such that $|a_n| < P$ for all n . Now

$$|a_n| = |a_n - A + A| \leq |a_n - A| + |A|.$$

But by hypothesis we can find N such that $|a_n - A| < \epsilon$ for all $n > N$, i.e.,

$$|a_n| < \epsilon + |A| \quad \text{for all } n > N.$$

It follows that $|a_n| < P$ for all n if we choose P as the largest one of the numbers $a_1, a_2, \dots, a_N, \epsilon + |A|$. $\square$

Theorem 4.4.5. *If $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$, then prove that $\lim_{n \rightarrow \infty} a_n b_n = AB$.*

Proof. We have, using theorem (4.4.4),

$$\begin{aligned} |a_n b_n - AB| &= |a_n (b_n - B) + B (a_n - A)| \\ &\leq |a_n| |b_n - B| + |B| |a_n - A| \\ &\leq P |b_n - B| + (|B| + 1) |a_n - A|. \end{aligned} \tag{4.4}$$

But since $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$, give any $\epsilon > 0$ find N_1 and N_2 such that

$$|b_n - B| < \frac{\epsilon}{2P} \quad \text{for all } n > N_1 \quad \text{and} \quad |a_n - A| < \frac{\epsilon}{2(|B| + 1)} \quad \text{for all } n > N_2.$$

Hence, from (4.4), $|a_n b_n - AB| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$ for all $n > N$, where N is the larger of N_1 and N_2 . Thus, the result is proved. $\square$

Theorem 4.4.6. *If $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$, then prove the following:*

$$(a) \quad \lim_{n \rightarrow \infty} \frac{1}{b_n} = \frac{1}{B},$$

$$(b) \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}.$$

Proof. (a) We must show that for any given $\epsilon > 0$, we can find N such that

$$\left| \frac{1}{b_n} - \frac{1}{B} \right| = \frac{|B - b_n|}{|B||b_n|} < \epsilon, \text{ for all } n > N. \quad (4.5)$$

By hypothesis, given any $\epsilon > 0$, we can find N_1 , such that $|b_n - B| < \frac{\epsilon}{2}B^2$ for all $n > N_1$.

Also, since $\lim_{n \rightarrow \infty} b_n = B \neq 0$, we can find N_2 such that $|b_n| > \frac{1}{2}|B|$ for all $n > N_2$.

Then if N is the larger of N_1 and N_2 , we can write (4.5) as

$$\left| \frac{1}{b_n} - \frac{1}{B} \right| = \frac{|B - b_n|}{|B||b_n|} < \frac{\frac{\epsilon}{2}B^2}{|B| \cdot \frac{1}{2}|B|} = \epsilon, \text{ for all } n > N,$$

and the proof is complete.

(b) From part (a) and theorem (4.4.5), we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \left(a_n \cdot \frac{1}{b_n} \right) \\ &= \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} \frac{1}{b_n} \\ &= A \cdot \frac{1}{B} = \frac{A}{B}. \end{aligned}$$

Hence, proof for part (b) is complete. □

4.5 Infinity, Bounded and Monotonic Sequences

4.5.1 Infinity

We write $\lim_{n \rightarrow \infty} a_n = \infty$ if for each positive number M we can find a positive number N (depending on M) such that $a_n > M$ for all $n > N$. Similarly, we write $\lim_{n \rightarrow \infty} a_n = -\infty$ if for each positive number M we can find a positive number N such that $a_n < -M$ for all $n > N$. It should be emphasized that ∞ and $-\infty$ are not numbers so that the

sequences are not convergent.

4.5.2 Bounded Sequence

If $u_n \leq M$ for $n = 1, 2, 3, \dots$, where M is a constant (independent of n), we say that the sequence $\{u_n\}$ is bounded above and M is called an upper bound. If $u_n \geq m$, the sequence is bounded below and m is called a lower bound.

If $m \leq u_n \leq M$ the sequence is called bounded. Often this is indicated by $|u_n| \leq P$.
Every convergent sequence is bounded, but the converse is not necessarily true.

4.5.3 Monotonic Sequence

If $u_{n+1} \geq u_n$ the sequence is called monotonic increasing; if $u_{n+1} > u_n$ it is called strictly increasing. Similarly, if $u_{n+1} \leq u_n$ the sequence is called monotonic decreasing, while if $u_{n+1} < u_n$ it is strictly decreasing.

Examples:

1. The sequence $1, 1.1, 1.11, 1.111 \dots$ is bounded and monotonic increasing. It is also strictly increasing.
2. The sequence $1, -1, 1, -1, 1, \dots$ is bounded but not monotonic increasing or decreasing.
3. The sequence $-1, -1.5, -2, -2.5, -3, \dots$ is monotonic decreasing and not bounded. However, it is bounded above.

Assignment: Prove that the sequence with n th term $u_n = \frac{2n-7}{3n+2}$ is (see, page 30, [Wrede and Spiegel \(2002\)](#))

- (a) monotonic increasing,
- (b) bounded above,

(c) bounded below,

(d) bounded, and

(e) has a limit.

Assignment: A sequence $\{u_n\}$ is defined by the recursion formula $u_{n+1} = \sqrt{3} u_n$, $u_1 = 1$ (see, page 30-31, [Wrede and Spiegel \(2002\)](#)).

(a) Prove that $\lim_{n \rightarrow \infty} u_n$ exists,

(b) Find the limit in (a).

Example:

Prove that the sequence with the n th term $u_n = \left(1 + \frac{1}{n}\right)^n$ is monotonic, increasing, bounded, and thus a limit exists.

Solution: If n is a positive integer, then by the binomial theorem we have

$$\begin{aligned} (1+x)^n &= 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots \\ &\quad + \frac{n(n-1)(n-2) \cdots (n-n+1)}{n!}x^n. \end{aligned}$$

Letting $x = \frac{1}{n}$,

$$\begin{aligned} u_n &= (1+x)^n \\ &= 1 + n \left(\frac{1}{n}\right) + \frac{n(n-1)}{2!} \left(\frac{1}{n}\right)^2 + \frac{n(n-1)(n-2)}{3!} \left(\frac{1}{n}\right)^3 + \dots \\ &\quad + \frac{n(n-1)(n-2) \cdots (n-n+1)}{n!} \left(\frac{1}{n}\right)^n. \\ &= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \dots \\ &\quad + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{n-1}{n}\right). \end{aligned}$$

Since each term beyond the first two terms in the last expression is an increasing function of n , it follows that the sequence u_n is a monotonic sequence.

It is also clear that

$$\begin{aligned}\left(1 - \frac{1}{n}\right)^n &< 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} \\ &< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}} < 3.\end{aligned}$$

Thus, u_n is bounded and monotonic increasing, and so has a limit.

4.6 Least Upper Bound and Greatest Lower Bound of a Sequence

4.6.1 Least Upper Bound of a Sequence

A number $\underline{M}$ is called the least upper bound (l.u.b.) of the sequence $\{u_n\}$ if $u_n \leq \underline{M}$, $n = 1, 2, 3, \dots$ while at least one term is greater than $\underline{M} - \epsilon$ for any $\epsilon > 0$.

4.6.2 Greatest Lower Bound of a Sequence

A number $\overline{m}$ is called the greatest lower bound (g.l.b.) of the sequence $\{u_n\}$ if $u_n \geq \overline{m}$, $n = 1, 2, 3, \dots$ while at least one term is less than $\overline{m} + \epsilon$ for any $\epsilon > 0$.

4.7 Limit Superior and Limit Inferior

A number $\bar{l}$ is called the limit superior, greatest limit or upper limit ($\limsup$ or $\overline{\lim}$) of the sequence $\{u_n\}$ if infinitely many terms of the sequence are greater than $\bar{l} - \epsilon$ while only a finite number of terms are greater than $\bar{l} + \epsilon$, where ϵ is any positive number.

A number $\underline{l}$ is called the limit inferior, least limit or lower limit ($\liminf$ or $\underline{\lim}$) of the sequence $\{u_n\}$ if infinitely many terms of the sequence are less than $\underline{l} + \epsilon$ while only a finite number of terms are less than $\underline{l} - \epsilon$, where ϵ is any positive number.

A sequence $\{u_n\}$ converges if and only if $\limsup (u_n) = \liminf (u_n)$.

Assignment: Find the $\limsup u_n$ and $\liminf u_n$ where

(a) $u_n = (-1)^n$,

(b) $u_n = \{4, 3.1, 5.9, 3.01, 5.99, 3.001, 5.999, \dots\}$, and

(c) $u_n = \frac{1}{n} + (-1)^n$.

Bolzano–Weierstrass Theorem

Every bounded monotonic (increasing or decreasing) sequence has a limit.

Proof: Suppose the given bounded infinite set is contained in the finite interval $[a, b]$. Divide this interval into two equal intervals. Then at least one of these, denoted by $[a_1, b_1]$, contains infinitely many points. Dividing $[a_1, b_1]$ into two equal intervals, we obtain another interval, say, $[a_2, b_2]$, containing infinitely many points. Continuing this process, we obtain a set of intervals $[a_n, b_n]$, $n = 1, 2, 3, \dots$ each interval contained in the preceding one and such that

$$\begin{aligned} b_1 - a_1 &= \frac{b - a}{2}, \\ b_2 - a_2 &= \frac{b_1 - a_1}{2} = \frac{b - a}{2^2} \\ &\vdots \\ b_n - a_n &= \frac{b - a}{2^n} \end{aligned}$$

from which we see that $\lim_{n \rightarrow \infty} (b_n - a_n) = \lim_{n \rightarrow \infty} \frac{b-a}{2^n} = 0$.

4.8 Infinite Series

Let $u_1, u_2, \dots$ be a given sequence. Form a new sequence $S_1, S_2, \dots$ where

$$S_1 = u_1$$

$$S_2 = u_1 + u_2$$

$$S_3 = u_1 + u_2 + u_3$$

$$\vdots$$

$$\begin{aligned} S_n &= u_1 + u_2 + u_3 + \cdots + u_n \\ &\vdots \end{aligned}$$

where S_n , called the n th partial sum, is the sum of first n terms of the sequence $\{u_n\}$.

The sequence $S_1, S_2, \cdots$ is symbolized by

$$u_1 + u_2 + u_3 + \cdots + u_n + \cdots = \sum_{i=1}^{\infty} u_i$$

which is called an infinite series. If $\lim_{n \rightarrow \infty} S_n = S$ exists, the series is called convergent and S is its sum, otherwise the series is called divergent.

Example:

Prove that the infinite series: $a + ar + ar^2 + ar^3 + \cdots = \sum_{n=1}^{\infty} ar^{n-1}$,

(a) converges to $\frac{a}{1-r}$ if $|r| < 1$, and

(b) diverges if $|r| \geq 1$.

Solution: First consider as

$$S_n = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1}. \quad (4.6)$$

$$rS_n = ar + ar^2 + ar^3 + \cdots + ar^{n-1} + ar^n. \quad (4.7)$$

Now subtract (4.7) from (4.6) and obtain as under

$$\begin{aligned} (1-r)S_n &= a - ar^n \\ \therefore S_n &= \frac{a(1-r^n)}{1-r}. \end{aligned}$$

(a) If $|r| < 1$, $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{a(1-r^n)}{1-r} = \frac{a}{1-r}$.

(b) If $|r| > 1$, $\lim_{n \rightarrow \infty} S_n$ does not exist.

Example:

Prove that if a series converges, its n th term must necessarily approach zero.

Solution: Since

$$S_n = u_1 + u_2 + \cdots + u_n \quad \text{and}$$

$$S_{n-1} = u_1 + u_2 + \cdots + u_{n-1}$$

we have

$$u_n = S_n - S_{n-1}.$$

If the series converges to S , then

$$\begin{aligned} \lim_{n \rightarrow \infty} u_n &= \lim_{n \rightarrow \infty} (S_n - S_{n-1}) \\ &= \lim_{n \rightarrow \infty} (S_n) - \lim_{n \rightarrow \infty} (S_{n-1}) \\ &= S - S \\ &= 0. \end{aligned}$$

Hence, if a series converges, its n th term must necessarily approach zero.

Chapter 5

Convergence

The convergence of sequences of random variables to some limit random variable is an important concept in probability theory, and its applications to statistics and stochastic processes. The same concepts are known in more general mathematics as stochastic convergence. There exist several modes of convergence of random variables, some of which are described below.

5.1 Convergence in Distribution

With this mode of convergence, we increasingly expect to see the next outcome in a sequence of random experiments becoming better and better modeled by a given probability distribution. Convergence in distribution is the weakest form of convergence. However, convergence in distribution is very frequently used in practice; most often it arises from application of the central limit theorem.

5.1.1 Definition

A sequence of real valued random variables $\{X_n\}$, $n = 1, 2, \dots$ is said to be **converge in distribution**, or **converge weakly**, or **converge in law** to a random variable X if it follows as

$$\lim_{n \rightarrow \infty} F_n(x) = F(x),$$

for every number $x \in \mathbb{R}$ at which F is continuous. Here F_n and F are the cumulative distribution functions of random variables X_n and X , respectively.

Convergence in distribution may be denoted as

$$X_n \xrightarrow{L} X \text{ or } X_n \xrightarrow{d} X.$$

5.1.2 Properties

Some of the properties of convergence in distribution are as follow:

1. Let $X_n \xrightarrow{L} X$ and c is a constant, then

(i) $X_n + c \xrightarrow{L} X + c$, and

(ii) $cX_n \xrightarrow{L} cX$.

2. $X_n \xrightarrow{L} X$ and $Y_n \xrightarrow{L} c$, then

(i) $X_n + Y_n \xrightarrow{L} X + c$,

(ii) $X_n Y_n \xrightarrow{L} cX$, and

(iii) $\frac{X_n}{Y_n} \xrightarrow{L} \frac{X}{c}$ ($c \neq 0$).

5.1.3 Examples

Example 1: Let $\{X_n\}$ be a sequence of binomial variates having distribution

$$Pr[X_n = r] = \binom{n}{r} P_n^r (1 - P_n)^{n-r}.$$

Let $E[X_n] = nP_n = \lambda$ be a finite constant, then

$$Pr[X_n = r] \rightarrow \frac{e^{-\lambda} \lambda^r}{r!} \text{ as } n \rightarrow \infty.$$

Example 2: Let $\{X_n\}$ be a sequence of t variates having distribution

$$f(X_n = t) = \frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{1}{\left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}}.$$

then it follows that

$$f(X_n = t) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}t^2} \text{ as } n \rightarrow \infty.$$

5.2 Convergence in Probability

The basic idea behind this type of convergence is that the probability of an “unusual” outcome becomes smaller and smaller as the sequence progresses. The concept of convergence in probability is used very often in statistics. *For example*, an estimator is called consistent if it converges in probability to the quantity being estimated. Convergence in probability is also the type of convergence established by the weak law of large numbers.

5.2.1 Definition

A sequence of random variables $\{X_n\}$, $n = 1, 2, \dots$ is said to be **converges in probability** towards the random variable X if for all $\epsilon > 0$

$$\lim_{n \rightarrow \infty} Pr(|X_n - X| > \epsilon) = 0.$$

Convergence in probability is denoted by adding the letter p over an arrow indicating convergence, or using the “plim” probability limit operator:

$$X_n \xrightarrow{p} X \text{ or } X_n \xrightarrow{P} X \text{ or } \text{plim}_{n \rightarrow \infty} X_n = X.$$

5.2.2 Properties

Some of the properties of convergence in probability are as follow:

1. $X_n \xrightarrow{P} X \Rightarrow X_n - X \xrightarrow{P} 0.$
2. $X_n \xrightarrow{P} X, X_n \xrightarrow{P} Y \Rightarrow Pr[X = Y] = 1.$
3. $X_n \xrightarrow{P} X, Y_n \xrightarrow{P} Y$, then

- (i) $X_n \pm Y_n \xrightarrow{P} X \pm Y$
- (ii) $X_n Y_n \xrightarrow{P} XY$
- (iii) $\frac{X_n}{Y_n} \xrightarrow{P} \frac{X}{Y}$
- 4. $X_n \xrightarrow{P} X$, and k is a constant $\Rightarrow kX_n \xrightarrow{P} kX$.
- 5. $X_n \xrightarrow{P} k \Rightarrow X_n^2 \xrightarrow{P} k^2$.
- 6. $X_n \xrightarrow{P} a, Y_n \xrightarrow{P} b \Rightarrow X_n Y_n \xrightarrow{P} ab$, where a, b are constants.
- 7. $X_n \xrightarrow{P} X$ and Y is a random variable $\Rightarrow X_n Y \xrightarrow{P} XY$.
- 8. $X_n \xrightarrow{P} 1 \Rightarrow X_n^{-1} \xrightarrow{P} 1$.
- 9. Convergence in probability implies convergence in distribution: $X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X$.

All the above statement can be easily varified.

5.2.3 Some Examples

Example 1: Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed random variables with $\mu = E(X_i)$ and $\sigma^2 = Var(X_i) < \infty$ for $i = 1, 2, \dots, \infty$. Then sample mean $\bar{X}_n$ converges in probability to μ .

Solution: We have the following

$$E(\bar{X}_n) = \mu \quad \text{and} \quad Var(\bar{X}_n) = \frac{\sigma^2}{n}.$$

By Chebyshev's inequality

$$Pr(|\bar{X}_n - E(\bar{X}_n)| \geq \epsilon) \leq \frac{Var(\bar{X}_n)}{\epsilon^2},$$

for $\epsilon > 0$. Hence

$$Pr(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}.$$

Taking the limit as n tends to infinity, we get

$$\lim_{n \rightarrow \infty} \Pr(|\bar{X}_n - \mu| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{\sigma^2}{n\epsilon^2},$$

which yields

$$\lim_{n \rightarrow \infty} \Pr(|\bar{X}_n - \mu| \geq \epsilon) = 0.$$

So the sample mean $\bar{X}_n$ converges in probability to μ , also written as $\bar{X}_n \xrightarrow{P} \mu$.

Example 2: Let $X_n \sim \text{Exponential}(n)$, show that $X_n \xrightarrow{P} 0$. That is, the sequence $X_1, X_2, \dots$ converges in probability to the zero random variable X .

Solution: We have

$$\begin{aligned} \lim_{n \rightarrow \infty} \Pr(|X_n - 0| \geq \epsilon) &= \lim_{n \rightarrow \infty} \Pr(X_n \geq \epsilon) \\ &= \lim_{n \rightarrow \infty} [1 - \Pr(X_n \leq \epsilon)] \\ &= \lim_{n \rightarrow \infty} [1 - (1 - e^{-n\epsilon})] \\ &= \lim_{n \rightarrow \infty} e^{-n\epsilon} \\ &= 0, \quad \text{for all } \epsilon > 0. \end{aligned}$$

Hence the sequence of random variables $X_1, X_2, \dots$ convergence in probability to the zero random variable X .

Example 3: If $X \sim \text{Binomial}(n, p)$, then $X_n \xrightarrow{P} p$.

Solution: **Assignment**

Theorem 5.2.1. *Convergence in probability implies convergence in distribution. That can be also written as*

$$X_n \xrightarrow{P} X \quad \Rightarrow \quad X_n \xrightarrow{d} X.$$

Proof. In order to prove convergence in distribution, one must show that the sequence of cumulative distribution functions converges to the F_X at every point where F_X is continuous. Let a be such a point. For every $\epsilon > 0$, we have

$$\begin{aligned}\Pr(X_n \leq a) &\leq \Pr(X \leq a + \epsilon) + \Pr(|X_n - X| > \epsilon) \\ \Pr(X \leq a - \epsilon) &\leq \Pr(X_n \leq a) + \Pr(|X_n - X| > \epsilon)\end{aligned}$$

So, we have

$$\begin{aligned}\Pr(X \leq a - \epsilon) - \Pr(|X_n - X| > \epsilon) &\leq \Pr(X_n \leq a) \\ &\leq \Pr(X \leq a + \epsilon) + \Pr(|X_n - X| > \epsilon).\end{aligned}$$

Taking the limit as $n \rightarrow \infty$, we obtain:

$$F_X(a - \epsilon) \leq \lim_{n \rightarrow \infty} \Pr(X_n \leq a) \leq F_X(a + \epsilon),$$

where $F_X = \Pr(X \leq a)$ is the cumulative distribution function of X . This function is continuous at a by assumption, and therefore both $F_X(a - \epsilon)$ and $F_X(a + \epsilon)$ converge to $F_X(a)$ as $\epsilon \rightarrow 0^+$. Taking this limit, we obtain

$$\lim_{n \rightarrow \infty} \Pr(X_n \leq a) = \Pr(X \leq a),$$

which means that $\{X_n\}$ converges to X in distribution. □

5.3 Almost Sure Convergence

This is the type of stochastic convergence that is most similar to pointwise convergence known from elementary real analysis.

5.3.1 Definition

A sequence of random variables $\{X_n\}$, $n = 1, 2, \dots$ is said to be converges **almost surely (a.s)** or **almost everywhere** or **with probability 1** or **strongly** to another

random variable X if and only if

$$Pr \left[\lim_{n \rightarrow \infty} X_n = X \right] = 1,$$

or, equivalently,

$$\lim_{N \rightarrow \infty} Pr \left[\sup_{n \geq N} |X_n - X| > \epsilon \right] = 0 \text{ for every } \epsilon.$$

Almost sure convergence is often denoted by adding the letters *a.s.* over an arrow indicating convergence: $X_n \xrightarrow{a.s.} X$ or $X_n \rightarrow X$ with probability 1.

5.3.2 Properties

Some of the properties of almost surely are as follow:

1. If $X_n \xrightarrow{a.s.} X$ and $X_n \xrightarrow{a.s.} Y$, then $X = Y$ almost surely.
2. If $X_n \xrightarrow{a.s.} X$ and $Y_n \xrightarrow{a.s.} Y$, then $aX_n + bY_n \xrightarrow{a.s.} aX + bY$, $\forall a, b \in \mathbb{R}$.
3. If $X_n \xrightarrow{a.s.} X$ and $Y_n \xrightarrow{a.s.} Y$, then $X_n Y_n \xrightarrow{a.s.} XY$.
4. Almost sure convergence implies convergence in probability, and hence implies convergence in distribution: $X_n \xrightarrow{a.s.} X \Rightarrow X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X$.
5. Convergence in probability implies there exists a sub-sequence (k_n) which almost surely converges: $X_n \xrightarrow{P} X \Rightarrow X_{k_n} \xrightarrow{a.s.} X$.

5.3.3 Examples

Example 1: Consider an animal of some short-lived species. We record the amount of food that this animal consumes per day. This sequence of numbers will be unpredictable, but we may be quite certain that one day the number will become zero, and will stay zero forever after.

Example 2: Consider a man who tosses seven coins every morning. Each afternoon, he donates one pound to a charity for each head that appeared. The first time the

result is all tails, however, he will stop permanently.

Let $X_1, X_2, \dots$ be the daily amounts the charity received from him. We may be almost sure that one day this amount will be zero, and stay zero forever after that.

Theorem 5.3.1. *Convergence almost surely implies convergence in probability. That can be also written as*

$$X_n \xrightarrow{a.s.} X \Rightarrow X_n \xrightarrow{P} X.$$

Proof. We have

$$|X_n - X| > \epsilon \Rightarrow \sup_{n \geq N} |X_n - X| > \epsilon.$$

So that

$$Pr[|X_n - X| > \epsilon] \leq Pr\left[\sup_{n \geq N} |X_n - X| > \epsilon\right].$$

According to the law of large numbers

$$Pr\left[\sup_{n \geq N} |X_n - X| > \epsilon\right] = 0 \text{ as } n \rightarrow \infty.$$

Therefore,

$$Pr[|X_n - X| > \epsilon] \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Hence, $X_n \xrightarrow{a.s.} X \Rightarrow X_n \xrightarrow{P} X.$

□

5.4 Laws of Large Numbers

The law of large numbers (*LLN*) is a theorem that describes the result of performing the same experiment a large number of times. According to the law, the average of the results obtained from a large number of trials should be close to the expected value, and will tend to become closer as more trials are performed.

The *LLN* is important because it guarantees stable long-term results for the averages

of some random events.

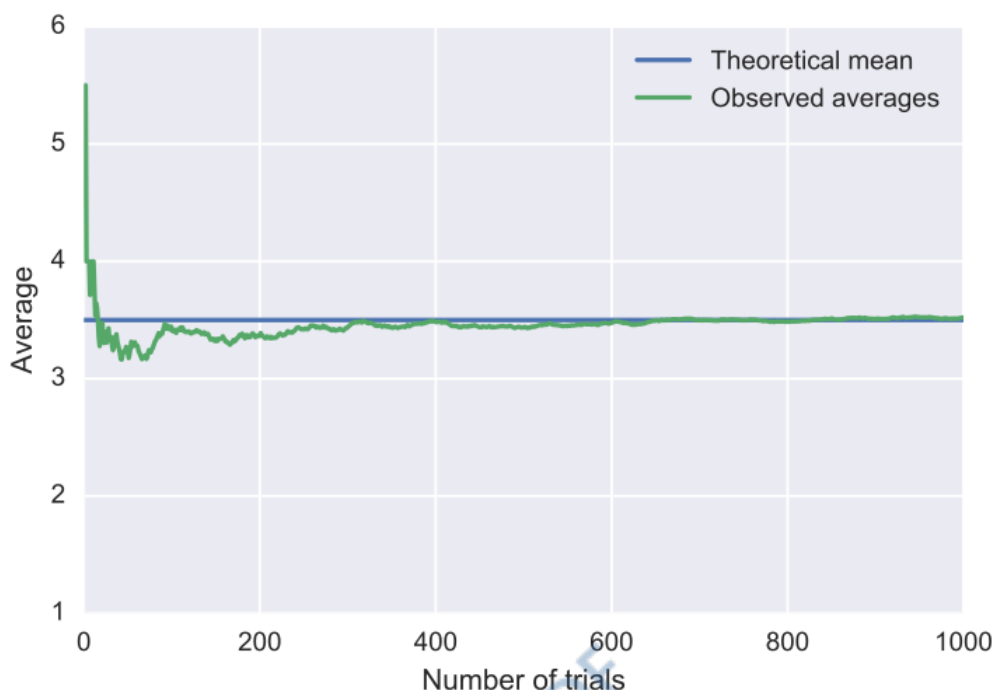


Figure 5.1: Average Dice Roll by Number of Rolls.

An illustration of the law of large numbers using a particular run of rolls of a single dice. As the number of rolls in this run increases, the average of the values of all the results approaches 3.5. While different runs would show a different shape over a small number of throws (at the left), over a large number of rolls (to the right) they would be extremely similar.

Two different versions of the law of large numbers are described below; they are called the **strong law of large numbers**, and the **weak law of large numbers**.

Chebyshev's Inequality: Let X be a random variable with mean μ and standard deviation σ . Then Chebyshev's inequality states that

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \quad \text{Or,} \quad P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2},$$

for any nonzero positive constant k .

5.4.1 Weak Law of Large Numbers

The weak law of large numbers (WLLN) states that the sample average converges in probability towards the expected value

$$\overline{X}_n \xrightarrow{P} \mu, \quad \text{when } n \rightarrow \infty.$$

That is, for any positive number ϵ ,

$$\lim_{n \rightarrow \infty} \Pr(|\overline{X}_n - \mu| > \epsilon) = 0.$$

Interpreting this result, the weak law states that for any nonzero margin specified, no matter how small, with a sufficiently large sample there will be a very high probability that the average of the observations will be close to the expected value; that is, within the margin.

Necessary and Sufficient Conditions

Necessary and sufficient conditions for the existence of WLLN are

- (i) $E(X_i)$ exists for all i ,
- (ii) $B_n = \text{Var}[X_1, X_2, \dots, X_n]$ exists and
- (iii) $\frac{B_n}{n^2} \rightarrow 0$ as $n \rightarrow \infty$.

Condition (i) is necessary without it the law itself cannot be stated. But the condition (ii) and (iii) are not necessary, (iii) is however a sufficient condition.

Theorem 5.4.1 (Chebyshev's Law of Large Numbers). *Given $X_1, X_2, \dots$ an infinite sequence of i.i.d. random variables with finite expected value $E(X_i) = \mu < \infty \forall i$, we are interested in the convergence of the sample average $\overline{X}_n = \frac{1}{n}(X_1 + \dots + X_n)$.*

The weak law of large numbers states:

$$\overline{X}_n \xrightarrow{P} \mu \quad \text{when } n \rightarrow \infty.$$

Proof. This proof uses the assumption of finite $\text{Var}(X_i) = \sigma^2$ (for all i). The independence of the random variables implies no correlation between them, and we have that as the following

$$\begin{aligned} \text{Var}(\overline{X}_n) &= \text{Var}\left(\frac{1}{n}(X_1 + \cdots + X_n)\right) \\ &= \frac{1}{n^2} \text{Var}(X_1 + \cdots + X_n) \\ &= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}. \end{aligned}$$

The common mean μ of the sequence is the mean of the sample average:

$$E(\overline{X}_n) = \mu.$$

Using Chebyshev's inequality on $\overline{X}_n$ results in

$$P(|\overline{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

This may be used to obtain the following:

$$P(|\overline{X}_n - \mu| < \varepsilon) = 1 - P(|\overline{X}_n - \mu| \geq \varepsilon) \geq 1 - \frac{\sigma^2}{n\varepsilon^2}.$$

As n approaches infinity, the expression approaches 1. And by definition of convergence in probability, we have obtained

$$\overline{X}_n \xrightarrow{P} \mu \quad \text{when } n \rightarrow \infty.$$

Hence the theorem. □

5.4.2 Strong Law of Large Numbers

The strong law of large numbers states that the sample average converges almost surely to the expected value

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu, \quad \text{when } n \rightarrow \infty.$$

That is,

$$\Pr\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

What this means is that the probability that, as the number of trials "n" goes to infinity, the average of the observations converges to the expected value, is equal to one.

Necessary and Sufficient Conditions

Necessary and sufficient conditions for the existence of SLLN are

(i) $E(X_i)$ exists for all i , and

(ii) $\sum_{i=1}^{\infty} \frac{\sigma_i^2}{i^2} < \infty$.

Condition (i) is necessary for the existence of strong law of large numbers and condition (ii) is sufficient. That is convergence of $\sum_{i=1}^{\infty} \frac{\sigma_i^2}{i^2}$ is sufficient for the existence of strong law of large numbers.

Theorem 5.4.2 (Kolmogorov Law of Large Numbers). *Let $\{X_i\}$, $i = 1, 2, \dots$, be a sequence of independent random variables such that $E(X_i) = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2$. Then hold the following*

$$\sum_{i=1}^{\infty} \frac{\sigma_i^2}{i^2} < \infty \quad \Rightarrow \quad \bar{X}_n - \bar{\mu}_n \xrightarrow{\text{a.s.}} 0,$$

that is, the sequence $X_1, X_2, \dots$ obeys the strong law of large numbers.

Proof. **Assignment □

Assignment: Differentiate between WLLN and SLLN.

Example: Let X_i assume two values i and $-i$ with equal probabilities. Show that the law of large numbers cannot be applied to the independent random variables $X_1, X_2, \dots$, i.e., X_i 's.

Solution: We have

$$P[X_i = i] = \frac{1}{2}, \quad P[X_i = -i] = \frac{1}{2}.$$

So we can obtain as

$$E[X_i] = \frac{1}{2}(i) + \frac{1}{2}(-i) = 0, \quad i = 1, 2, \dots,$$

and

$$Var[X_i] = E[X_i^2] = \frac{i^2}{2} + \frac{i^2}{2} = i^2, \quad i = 1, 2, \dots.$$

Since $X_1, X_2, \dots, X_n$ are independent random variables, we can obtain as

$$\begin{aligned} \frac{B_n}{n^2} &= \frac{1}{n^2} Var[X_1 + X_2 + \dots + X_n] \\ &= \frac{1}{n^2} [1^2 + 2^2 + \dots + n^2] \\ &= \frac{n(n+1)(2n+1)}{6n^2} \\ &= \frac{n(1 + \frac{1}{n})(2 + \frac{1}{n})}{6} \rightarrow \infty \text{ as } n \rightarrow \infty. \end{aligned}$$

Hence law of large numbers does not hold.

Example: Let X_i can have two values i^α and $-i^\alpha$ with equal probabilities. Show that the law of large numbers can be applied to the independent random variables $X_1, X_2, \dots$, if $\alpha < \frac{1}{2}$.

Solution: We have

$$P[X_i = i^\alpha] = \frac{1}{2}, \quad P[X_i = -i^\alpha] = \frac{1}{2}.$$

So we can obtain as

$$E[X_i] = \frac{1}{2}(i^\alpha) + \frac{1}{2}(-i^\alpha) = 0, \quad i = 1, 2, \dots,$$

and

$$\text{Var}[X_i] = E[X_i^2] = \frac{(i^\alpha)^2}{2} + \frac{(-i^\alpha)^2}{2} = i^{2\alpha}, \quad i = 1, 2, \dots.$$

Since $X_1, X_2, \dots, X_n$ are independent random variables, we can obtain as

$$\begin{aligned} \frac{B_n}{n^2} &= \frac{1}{n^2} \text{Var}[X_1 + X_2 + \dots + X_n] \\ &= \frac{1}{n^2} [1^{2\alpha} + 2^{2\alpha} + \dots + n^{2\alpha}] \\ &= \frac{1}{n^2} \int_0^n x^{2\alpha} dx \quad [\text{From Euler-Maclaurin's Formula}] \\ &= \frac{1}{n^2} \left[\frac{x^{2\alpha+1}}{2\alpha+1} \right]_0^n \\ &= \frac{1}{n^2} \cdot \frac{n^{2\alpha+1}}{2\alpha+1} \\ &= \frac{n^{2\alpha-1}}{2\alpha+1} \rightarrow 0 \text{ as } n \rightarrow \infty \text{ if } \alpha < \frac{1}{2}. \end{aligned}$$

Hence the results follows law of large numbers.

Example: Let $\{X_n\}$ be mutually independent and identically distributed random variables with mean μ and finite variance. If $S_n = X_1 + X_2 + \dots + X_n$, prove that the law of large numbers does not hold for the sequence $\{S_n\}$.

Solution: Since $S_1, S_2, \dots, S_n$ are mutually independent and identically distributed

random variables, so

$$\begin{aligned}
 \frac{B_n}{n^2} &= \frac{1}{n^2} \text{Var} [S_1 + S_2 + \cdots + S_n] \\
 &= \frac{1}{n^2} \text{Var} [X_1 + (X_1 + X_2) + \cdots + (X_1 + X_2 + \cdots + X_n)] \\
 &= \frac{1}{n^2} \text{Var} [nX_1 + (n-1)X_2 + \cdots + 2X_{n-1} + X_n] \\
 &= \frac{1}{n^2} [n^2 \text{Var} (X_1) + (n-1)^2 \text{Var} (X_2) + \cdots + 2^2 \text{Var} (X_{n-1}) + 1^2 \text{Var} (X_n)]
 \end{aligned}$$

Let $\text{Var} (X_i) = \sigma^2$ for all i , therefore

$$\begin{aligned}
 \frac{B_n}{n^2} &= \frac{\sigma^2}{n^2} [1^2 + 2^2 + \cdots + n^2] \\
 &= \frac{\sigma^2 n(n+1)(2n+1)}{6n^2} \\
 &= \frac{\sigma^2 n(1 + \frac{1}{n})(2 + \frac{1}{n})}{6} \rightarrow \infty \text{ as } n \rightarrow \infty.
 \end{aligned}$$

Hence law of large numbers does not hold for the sequence $\{S_n\}$.

5.5 Central Limit Theorem

The central limit theorem (CLT) establishes that, in some situations, when independent random variables are added, their properly normalized sum tends toward a normal distribution (informally a “bell curve” distribution) even if the original variables themselves are not normally distributed. The theorem is a key concept in probability theory because it implies that probabilistic and statistical methods that work for normal distributions can be applicable to many problems involving other types of distributions.

For example, suppose that a sample is obtained containing a large number of observations, each observation being randomly generated in a way that does not depend on the values of the other observations, and that the arithmetic mean of the observed values is computed. If this procedure is performed many times, the central limit theorem says that the distribution of the average will be closely approximated

by a normal distribution. A simple example of this is that if one flips a coin many times the probability of getting a given number of heads in a series of flips will approach a normal curve, with mean equal to half the total number of flips in each series.

This theorem was first stated by Laplace in 1812 and a rigorous proof under fairly general conditions was given by Liapounoff in 1901. Below list some particular cases of this general central limit theorem.

- (i) De-Moivre's Laplace theorem,
- (ii) Lindeberge-Levy theorem,
- (iii) Liapounov's theorem, and
- (iv) Lindeberg-Feller theorem.

5.5.1 Classical Central Limit Theorem

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample of size n , that is, a sequence of independent and identically distributed random variables drawn from a distribution of expected value given by μ and finite variance given by σ^2 . Suppose we are interested in the sample average

$$S_n = \frac{X_1 + \dots + X_n}{n}$$

of these random variables. By the law of large numbers, the sample averages converge in probability and almost surely to the expected value μ as $n \rightarrow \infty$. The classical central limit theorem describes the size and the distributional form of the stochastic fluctuations around the deterministic number μ during this convergence. More precisely, it states that as n gets larger, the distribution of the difference between the sample average S_n and its limit μ , when multiplied by the factor $\sqrt{n}$ (that is $\sqrt{n}(S_n - \mu)$), approximates the normal distribution with mean 0 and variance σ^2 . For large enough n , the distribution of S_n is close to the normal distribution with mean μ and variance

$\frac{\sigma^2}{n}$. The usefulness of the theorem is that the distribution of $\sqrt{n}(S_n - \mu)$ approaches normality regardless of the shape of the distribution of the individual X_i . Formally, the theorem can be stated as follows:

Theorem 5.5.1 (Lindeberg–Lévy CLT). *Suppose $\{X_1, X_2, \dots\}$ is a sequence of independent and identically distributed random variables with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ convergence in distribution to a normal $N(0, \sigma^2)$:*

$$\sqrt{n}(S_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

Proof. The central limit theorem has a simple proof using characteristic function. It is similar to the proof of the (weak) law of large numbers.

Assume $\{X_1, X_2, \dots, X_n\}$ are independent and identically distributed random variables, each with mean μ and finite variance σ^2 . The sum $X_1 + X_2 + \dots + X_n$ has mean $n\mu$ and variance $n\sigma^2$. Consider the random variable

$$Z_n = \frac{X_1 + \dots + X_n - n\mu}{\sqrt{n\sigma^2}} = \sum_{i=1}^n \frac{X_i - \mu}{\sqrt{n\sigma^2}} = \sum_{i=1}^n \frac{1}{\sqrt{n}} Y_i,$$

where in the last step we defined the new random variables $Y_i = \frac{X_i - \mu}{\sigma}$, each with zero mean and unit variance. The characteristic function of Z_n is given by

$$\varphi_{Z_n}(t) = \varphi_{\sum_{i=1}^n \frac{1}{\sqrt{n}} Y_i}(t) = \varphi_{Y_1}\left(\frac{t}{\sqrt{n}}\right) \varphi_{Y_2}\left(\frac{t}{\sqrt{n}}\right) \cdots \varphi_{Y_n}\left(\frac{t}{\sqrt{n}}\right) = \left[\varphi_{Y_1}\left(\frac{t}{\sqrt{n}}\right)\right]^n,$$

where in the last step we used the fact that all of the Y_i are identically distributed.

The characteristic function of Y_1 is, by Taylor's theorem,

$$\varphi_{Y_1}\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right), \quad \left(\frac{t}{\sqrt{n}}\right) \rightarrow 0$$

where $o(t^2)$ is little o notation for some function of t that goes to zero more rapidly than t^2 . By the limit of the exponential function ($e^x = (1 + \frac{x}{n})^n$), the characteristic

function of Z_n equals

$$\varphi_{Z_n}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right)^n \rightarrow e^{-\frac{1}{2}t^2}, \quad n \rightarrow \infty.$$

All of the higher order terms vanish in the limit $n \rightarrow \infty$. The right hand side equals the characteristic function of a standard normal distribution $N(0, 1)$, which implies through Lévy's continuity theorem that the distribution of Z_n will approach $N(0, 1)$ as $n \rightarrow \infty$. Therefore, the sum $X_1 + \cdots + X_n$ will approach that of the normal distribution $N(n\mu, n\sigma^2)$, and the sample average

$$S_n = \frac{X_1 + \cdots + X_n}{n}$$

converges to the normal distribution, $N\left(\mu, \frac{\sigma^2}{n}\right)$. □



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